Article

AMERICAN MARKETING ASSOCIATION

Journal of Marketing Research
2026, Vol. 63(1) 167-190
© American Marketing Association 2025
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/00222437251373042
journals.sagepub.com/home/mrj
Sage

When Words Meet Visuals: How Content Composition Drives Social Media Engagement for Marketer-Generated Content

Ivy Chu Dang, Canice M.C. Kwan, Jayson S. Jia, and Yang Shi

Abstract

How does the balance between text and pictorial content in marketer-generated social media posts affect user engagement? The authors address this question by using computer vision and natural language processing tools to extract the visual and textual features of 34,610 organic brand posts from Facebook and Instagram. Using a confounding-and-cluster-robust causal forests model, they test how the balance of text and picture affects social media engagement across content and visual contexts. Results show that posts with greater emphasis on overlay text over pictorial content tend to have fewer likes and comments. However, the performance of text-oriented posts improves if text is more centered, informative, emotionally positive, and congruent with the pictorial content and if the picture contains fewer prominent objects or less information such as social cues. They quantify how incremental changes in such content composition affect social media engagement. These findings set forth evidence-based principles for optimizing text and picture balance in marketer-generated content and provide actionable guidelines on whether, where, when, and how to present text on an image. This research highlights the potential for transforming content and media creation from an imprecise art form into an empirical science nested within a data-driven visual optimization framework.

Keywords

text–picture balance, text and image, social media, marketer-generated content, visual marketing, marketing communications, causal machine learning, content optimization

Online supplement: https://doi.org/10.1177/00222437251373042

Submitted March 9, 2021

Written language and pictorial representations are the two most basic forms of human communications. Advertisers and artists alike have long sought to combine the strengths of each to maximize the impact of visual media (Ivins 1969; Smith 1991). Pictorial content is more engaging by virtue of being more eye-catching and easier to process cognitively, but text remains indispensable for conveying complex information—particularly marketing-mix information about brands, products, and offerings—that is difficult to communicate without written language (Wedel and Pieters 2017; Wyer, Hung, and Jiang 2008). The coexistence of text and picture,[1] while essential for effective communication, is not always harmonious: One element can distract or even detract from the other. This tension has received attention since at least Henry James (1893), who lamented that the rise of visual realism in the late 19th century—exemplified by greater emphasis on photorealistic illustrations in magazines and enriched stage designs for theatrical productions—threatened to overshadow the importance of verbal content in such media.

Today, the challenge of balancing text and picture persists in the production of marketing media and communications, particularly in digital and social media contexts. Platforms, ad slots,

[1] Throughout this article, we use the term "image(s)" to refer to the image file(s) attached to a post, which may include one or more images. "Picture" refers to the overall pictorial content within a post image, encompassing all nontextual visual content within the entire image.

Ivy Chu Dang (corresponding author) is Assistant Professor of Marketing, HKU Business School, The University of Hong Kong, Hong Kong (email: ivydang@hku.hk). Canice M.C. Kwan (corresponding author) is Assistant Professor of Marketing, Tourism, and Analytics, Nottingham University Business School, University of Nottingham, UK (email: canice.kwan@nottingham.ac.uk). Jayson S. Jia is Associate Professor of Marketing, HKU Business School, The University of Hong Kong, Hong Kong (email: jjia@hku.hk). Yang Shi is Associate Professor of Marketing, Shenzhen Audencia Financial Technology Institute, Shenzhen University, China (email: yolandashiy@connect.ust.hk).



Figure 1. Illustration of the Primary Components of a Social Media Post.
Notes: This figure presents a typical example of MGC on social media, featuring a post image displayed within a fixed, platform-specified frame. Within this frame, the content consists of overlay text and picture. The size of each component can be adjusted within the frame: Enlarging or shrinking one decreases or increases the relative salience of the other, thereby altering text–picture balance. The original post is displayed on the left, while its key components are visually illustrated on the right.

and devices impose fixed spatial parameters for content, so enlarging and emphasizing one element will necessarily deemphasize the other (see Figure 1). This creates an unavoidable trade-off in every post or advertisement: Allocating more space to pictorial content in an image might enhance visual appeal and capture initial attention, but it risks detracting from essential text-based information, such as product attributes, price, or promotional details. Conversely, prioritizing text by enlarging it might abet verbal processing but at the expense of capturing consumer attention (Kwan, Dai, and Wyer 2017; Pieters and Wedel 2004; Pillai, Katsikeas, and Presi 2012).

Despite text and picture being fundamental components of visual media, there is limited quantitative understanding of how their interplay affects consumer engagement. Previous research has primarily examined the relative merits of text versus picture. Marketers have long known that pictorially oriented ads are "better" than text-oriented ads at winning consumer attention (Childers and Houston 1984; Edell and Staelin 1983; Hartmann et al. 2021; Li and Xie 2020; Pieters and Wedel 2004; Rossiter and Percy 1997; Shepard 1967; Smith 1991). However, in practical scenarios, marketers and content creators do not face "either-or" decisions. Instead, they must strategically determine the optimal balance between text and picture that maximizes marketing effectiveness.

We use computer vision and natural language processing (NLP) tools to extract and quantify the relative balance of text and picture based on the proportion of their surface area, which we operationalize as text–image ratio (TIR), as well as other visual and text content features of 34,610 organic brand posts across nine major product categories from Facebook and Instagram. We then apply a machine-learning-based causal inference model, confounding-and-cluster-robust causal forests (CCR-CF), to investigate several open questions: First, we explore how the relative balance of text and pictorial content helps or hinders social media engagement for marketer-generated content (MGC), as measured by likes and comments. Second, we document several visual aesthetic and information-content moderators. In other words, we find that text and picture do not have a zero-sum relationship. Rather, the spatial arrangement of text and picture, their information content, and their content congruency can amplify or mitigate the main effect of TIR. Finally, we quantify and compare the effects of these different variables on social media engagement, highlighting the potential of applying data-driven visual content optimization in the marketplace.

Our research, a large-scale field study, differs in purpose, methodology, and scope from previous consumer behavioral research on how the visual features of ads affect consumer response. Most existing studies rely on lab experiments and online surveys (e.g., Kwan, Dai, and Wyer 2017; Pillai, Katsikeas, and Presi 2012), or in some cases eye tracking (Pieters and Wedel 2004; Pieters, Wedel, and Batra 2010; Pieters, Wedel, and Zhang 2007), to examine the psychological and visual mechanisms underlying consumer attention to ads. However, these approaches are inherently constrained by cost

and sample size limitations, typically focusing on a limited number of independent variables and relying on less diverse stimuli that may not adequately capture the diversity and complexity of contemporary marketing communications. Here, we leverage the latest computer vision toolkits to objectively and accurately quantify a host of visual features (e.g., TIR, object presence, color hue) for tens of thousands of brand posts, which we link to their social media engagement. Although our empirical analyses cannot directly reveal the underlying mechanisms, we use causal forest models to quantify the impact of text–picture balance and other variables on actual consumer response to MGC in the marketplace. This approach enables us to estimate the average treatment effect of TIR, uncover heterogeneous treatment effects across visual and content moderators, and control for a complex array of other factors.

Additionally, our research differs from recent empirical research that applies computer vision tools to analyze the visual properties of social media posts (e.g., Dzyabura, El Kihal, and Peres 2022; Li and Xie 2020; Liu, Dzyabura, and Mizik 2020; Zhang and Luo 2023). These studies have primarily focused on user-generated content (UGC), which differs from MGC in strategic purpose, content, design, and consumer interaction norms (Kumar et al. 2016; Meire et al. 2019). In our research, each moderator represents a common design consideration for marketing media: whether and where to place overlay text, what information to include, and how to visually present both text and picture. By quantifying how such considerations affect social media engagement, our findings provide marketers with actionable, data-driven insights on optimizing visual content.

Background

Balancing Text and Picture in MGC

MGC—firms' social media communications and content that are posted through their owned channels—is a key tool for driving consumer engagement with brands and plays a crucial role in the modern marketing funnel and integrated marketing communications (Goh, Heng, and Lin 2013; Lu, Dinner, and Grewal 2023; Meire et al. 2019). Consequently, nuances in the visual design and information content of MGC have become important managerial decisions, with implications for sales and brand perception that go beyond merely creating awareness or buzz (Liadeli, Sotgiu, and Verlegh 2023; Mochon et al. 2017).

A growing body of research employs advanced computer vision techniques to study how the visual properties of social media posts affect consumer response. Much of this work focuses on UGC, for example, by investigating how the visual content of users' posts affect engagement (e.g., Ceylan, Diehl, and Proserpio 2024; Dzyabura, El Kihal, and Peres 2022; Hartmann et al. 2021; Li and Xie 2020; Liu, Dzyabura, and Mizik 2020). However, UGC and MGC serve distinct roles in social media, and thus even identical content may elicit different responses when posted by consumers versus marketers (Goh, Heng, and Lin 2013). As a form of official brand communications, MGC is designed to be informative, persuasive, and interactive. For example, MGC may provide information about products, promotions, and events or encourage dialogue and discussion among followers or customers (Kumar et al. 2016; Lee, Hosanagar, and Nair 2018; Meire et al. 2019; Tellis et al. 2019).

When designing MGC, marketers face an inherent trade-off between the informativeness of text and the engagement power of picture: Text is essential for conveying complex concepts and granular technical information (e.g., product specification, statistics) and facilitates rational persuasion; in contrast, pictorial content is more effective at capturing attention, easier to process cognitively, and more memorable (Edell and Staelin 1983; MacInnis and Price 1987; see Wyer, Hung, and Jiang 2008 for a review). Previous research comparing the relative virtue of text versus pictorial content has predominantly relied on laboratory experiments to delineate cognitive and visual mechanisms behind how consumers process such content. Rooted in traditional advertising research, these studies typically consider a limited set of predictor variables that are examined in controlled settings (see Table 1 for a review). Most previous literature considers text and picture in discrete "mere presence" terms, employing experimental paradigms that compare content that is text only, picture only, or a combination of both. While decomposing the individual effects of text and picture provides psychological insight, this approach comes at the cost of external validity: In the marketplace, marketers rarely face a binary choice between text and picture but must instead determine the optimal balance between the two.

More recent empirical research combines computer vision and text analysis tools to analyze large samples of UGC content drawn from social media posts and consumer reviews. These studies have also focused on demonstrating "mere presence" effects, such as how user engagement benefits from more, higher-quality, or more relevant images (Ceylan, Diehl, and Proserpio 2024; Li and Xie 2020). They treat the presence of text in post images as a control feature rather than the focal construct and independent variable (e.g., Ahmadi, Waltenrath, and Janze 2023). In contrast, our research investigates the relationship between the relative balance of text and pictorial content and social media engagement.

Measuring and geometrically quantifying the visual features of such content at scale was challenging and costly before the off-the-shelf availability of pretrained computer vision toolkits, which we use. Only a few studies have directly examined the impact of the geometric size (i.e., surface area) of text or picture on consumer attention (e.g., Pieters and Wedel 2004; Pieters, Wedel, and Zhang 2007). Pieters and Wedel (2004) use eye-tracking analysis to show that in print ads, pictorial elements capture more baseline attention ("the picture superiority effect"), but attention levels do not change with the size of pictorial elements. They further show that text attracts attention proportionally to its surface area; increasing the surface area of text elements can boost ad attention by shifting focus

Table 1. Literature Review on Related Visual and Content Studies for Print Ads, UGC, and MGC.

				Text–Picture Effects			Visual and Content Contexts				
Study	**Data Source**	**Content Type**	**Outcome Variables**	**Inclusion of Image**	**Mere Presence of Text**	**Relative Balance of Text and Picture**	**Visual Aesthetic Features**	**Information Value of Text**	**Information Value of Picture**	**Content Congruency**	**Heterogeneous Text–Picture Effects Across Visual/Content Contexts**
This study	**Field (Facebook and Instagram)**	**MGC**	**Engagement (likes/comments)**	—	**Overlay text**	**Size**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Pieters, Wedel, and Zhang (2007)	Eye tracking	Print ads	Attention	—	—	Size	Yes	Yes	Yes	—	Yes
Pieters and Wedel (2004)	Eye tracking	Print ads	Attention	—	—	Size	—	Yes	Yes	—	Yes
Ahmadi, Waltenrath, and Janze (2023)	Field (Facebook and X)	MGC	Engagement shares	Presence	Overlay text (COV)	—	—	Yes	Yes	Yes	—
Zhang and Luo (2023)	Field (Yelp)	UGC (reviews)	Business survival	Number	Caption	—	Yes	Yes	Yes	—	—
He, Li, and Wang (2023)	Field (Airbnb)	MGC	Sales (booking rate)	Number (COV)	—	—	Yes	Yes	Yes	—	—
Lee, Hosanagar, and Nair (2018)	Field (Facebook)	MGC	Engagement (likes/comments)	Presence	—	—	—	Yes	—	—	—
Li and Xie (2020)	Field (Facebook and X)	UGC (posts)	Engagement (likes/shares)	Presence	—	—	Yes	Yes	Yes	Yes	—
Ceylan, Diehl, and Proserpio (2024)	Field (Yelp) + Experiment	UGC (reviews)	Review helpfulness (votes/likes)	Number (COV)	—	—	Yes	Yes	—	Yes	—
Kübler et al. (2024)	Field (Amazon)	UGC (reviews)	Review helpfulness (likes)	Number	—	—	—	Yes	—	—	—
Stafford (1996)	Lab experiment	Print ads	Memory and attitude	Presence	—	—	—	Yes	—	—	Yes
Edell and Staelin (1983)	Lab experiment	Print ads	Memory, attitude, belief, and purchase intention	Presence	—	—	—	Yes	—	Yes	Yes
Smith (1991)	Lab experiment	Print ads	Belief	Presence	—	—	—	—	—	Yes	Yes
Overgoor et al. (2022)	Field (Instagram)	MGC	Engagement (likes)	—	—	—	Yes	Yes	Yes	—	—
Hartmann et al. (2021)	Field (X and Instagram) and experiment	UGC (posts) and display ad	Engagement (likes/comments/clicks) and purchase intention	—	—	—	Yes	Yes	Yes	—	—
Jalali and Papatla (2019)	Field (X)	MGC	Engagement (shares)	—	—	—	—	Yes	—	—	—

Notes: "COV" denotes covariate.

between elements. In other words, emphasizing overlay text not only affects consumer attention to information conveyed in text but may also draw attention toward or away from other elements in a post. However, the precise impact of text size on visual attention depends on content. Pieters, Wedel, and Zhang (2007) find that text containing marketing-mix information like brand, price, and promotional elements receives outsized attention, with the effects varying for branded versus unbranded products. These findings suggest that the relative benefit of emphasizing text or picture depends on content and context, and that there are complex "it depends" rules beyond "mere presence" main effects. Subsequently, we discuss potential moderators, including the spatial arrangement of text, information value of content, and content congruency.

Text–Picture Balance on MGC

MGC on social media, like print media, integrates both pictorial and text information. However, unlike print media, consumer engagement with MGC is publicly observable and easily quantifiable. We analyzed social media posts from Instagram (2 billion monthly active users), which is relatively more pictorially oriented and experiential, and from Facebook (2.96 billion monthly active users), which is relatively more text oriented and informational. Across these and other social media platforms, all posts contain two basic content bins (Figure 1): (1) the image file (i.e., post image, which may or may not include text) and (2) the caption box (with a character limit of 2,200 on Instagram and 63,206 on Facebook). In our dataset, 72% of Facebook and 62% of Instagram posts include text within the post image, which publishers call "overlay text." Facebook posts contained a mean overlay text word count of 20.0 (SD = 34) and a mean caption text word count of 68.1 (SD = 76.1), and Instagram posts contained a mean overlay text word count of 15.5 (SD = 39.1) and a mean caption text word count of 53.3 (SD = 67.4). Despite platform differences, the use of overlay text remains prevalent, as in print media.

We operationalize the relative balance between text and pictorial content in the post image using TIR: the proportion of surface area allocated to text over the total image area. This definition is grounded in previous behavioral research (e.g., Kwan, Dai, and Wyer 2017; Pieters and Wedel 2004; Pieters, Wedel, and Zhang 2007) and treats the post image as a gestalt visual entity. Specifically, we define picture as encompassing all nontext content within an image, making text and picture inherently interdependent. Under this framework, a 1% increase in text size naturally corresponds to a 1% decrease in picture size, which is why we conceptualize text and picture size changes on a relative basis (see Web Appendix A). TIR offers a parsimonious and standardized means of assessing the relative distribution of text and pictorial content that is robust across content contexts and produces multifaceted effects.

There are several conceptual and technical reasons why we opted to use a holistic measurement of image-surface area instead of trying to measure (the sum of) the surface area of all individual pictorial content in MGC. First, extant theory is ambiguous in defining what qualifies as pictorial content: There is no standardized method to distinguish foreground from background (such as the burger and other components in the restaurant scene in Figure 1) or to determine whether abstract designs, styles, patterns, and intentional use of negative space (i.e., empty or less cluttered areas) constitute meaningful pictorial content. Second, there are technical limitations: Machine vision toolkits such as Google Vision AI are primarily trained to detect (and measure) marketing-relevant objects such as product images, brand logos, and human faces and often overlook background, decorative, partial, obscured, and visual-contextual content items that play a significant role in consumer visual perception. Third, text and picture often overlap in post images, which can generate misleading measurements of "total surface area of objects." Finally, consumers often derive the visual meaning of MGC from the holistic image during visual processing, particularly for product categories where the product is experiential, symbolic, or does not involve a tangible focal object. We discuss these issues in greater depth in Web Appendix A.

Content Moderators

Our primary analysis tests the main effect of TIR on social media engagement. The logic of the "picture superiority effect," established in previous visual research (Li and Xie 2020; Pieters and Wedel 2004), suggests that greater emphasis on text (i.e., higher TIR) could reduce engagement. To test whether this effect varies across visual and content contexts, we consider four categories of moderators: spatial arrangement of text (centeredness), information value of text (text entropy, text sentiment), information value of picture (visual entropy, social content), and content congruency (text–picture congruency; see Figure 2 for examples). Each reflects a common design consideration: where to insert text, what information to include, and how to present text. Figure 3 shows our conceptual framework.

Spatial arrangement of text. The spatial arrangement or relative position of the overlay text may moderate the effect of TIR. Eye-tracking research shows that when consumers view ads, their initial visual fixation is usually at the center of the ads (Rayner et al. 2001). Furthermore, consumers typically fixate on content in the center of ads more frequently and for longer (see Wedel and Pieters [2017] for a review). Consequently, the centeredness of the overlay text should affect its visual salience (relative to the pictorial content) and, by extension, how much attention and information processing the text content receives. Assuming that the text content of a post contains compelling information, we expect posts to perform better if the overlay text is placed at the center of the post image.

Information value of text. We expect the information value of the text to moderate the effect of TIR. Larger text is more legible and easier to cognitively process. Thus, emphasizing the text by enlarging it (i.e., higher TIR) should improve engagement when overlay text contains more information content and useful marketing-relevant information in particular. Indeed, Pieters, Wedel, and Zhang (2007) find that the surface size of marketing-mix

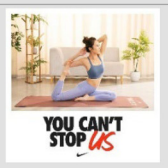
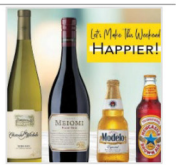
	Text–Picture Balance and Spatial Arrangement	Information Value of Text		Information Value of Picture		Content Congruency
	Text–image ratio and centeredness	Text entropy	Text sentiment	Visual entropy	Social content	Text–picture congruency
High						 Text: "A good appetite keeps your child strong inside"
	TIR: 96% Centeredness: 0	Text entropy: 2.8 (TIR: 44%)	Text sentiment: .91	Visual entropy: 2.3	Social content: 1	Congruency: .48
Medium						 Text: "Let's Make This Weekend Happier!"
	TIR: 40% Centeredness: −.17	Text entropy: 2 (TIR: 44%)	Text sentiment: 0	Visual entropy: 1.61	Social content: 1	Congruency: .37
Low						 Text: "vivo MILAN"
	TIR: 4% Centeredness: −.28	Text entropy: 0 (TIR: 43%)	Text sentiment: −.76	Visual entropy: 0	Social content: 1	Congruency: .19

Figure 2. Illustrative Examples of Text–Picture Balance and Content Moderators.
Notes: These examples are real-world MGC drawn from our data sample, featuring varying values (high, medium, and low).

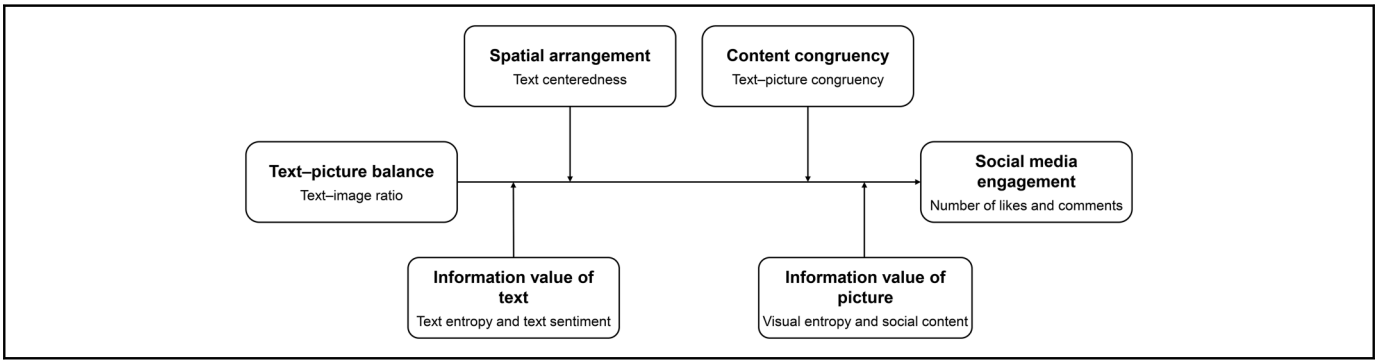


Figure 3. Conceptual Framework.

information such as price and promotion content in feature ads has an outsized impact on consumer attention and engagement.

We further consider the affective sentiment of text content as a moderator. Marketing content often contains emotional text, which can boost social media engagement, particularly if the text evokes strong positive emotions (Berger and Milkman 2012). Although the sentiment of content does not necessarily denote greater informativeness, emotional text can amplify, contextualize, or even shape the message of pictures (Wyer, Hung, and Jiang 2008), thereby changing their information value. Posts with pictorial content often benefit from supporting affective text. Making such text larger may enhance comprehension and engagement.

Information value of picture. We expect a negative moderation of the effect of TIR when the pictorial content of a post image conveys more information. When pictures contain more

Table 2. Summary of Data Processing Tools and Variables Extracted.

Original Post	Raw Content Extracted	Data Processing	Variable Extracted
	**Raw post image**  **Raw caption text** Our in-depth digital tools can give you a clearer view of how your income sources factor into your retirement lifestyle. Discuss your outlook with a financial advisor to see how our one-on-one guidance and financial solutions can help you realize your retirement vision.	**Google Vision AI** • OCR detection • Logo detection • Object detection • Face detection	**Content (marked in raw post image at left)** • Raw overlay text • Brand logos • Visual entropy (objects) • Social content (human faces)
		Google Vision AI • Image properties	**Visual aesthetics** • Visual complexity • Colorfulness • Color hues • Brightness • Brightness contrast
		Google Natural Language AI • Entity analysis • Sentiment analysis	• Text entropy (overlay text) • Marketing-mix information (overlay text) • Text sentiment (overlay text)
		Open AI's CLIP	• Text–picture congruency
		Google Natural Language AI • Entity analysis • Sentiment analysis	• Text entropy (caption) • Text sentiment (caption)

information, there is relatively less need for text, which might introduce redundant information. Furthermore, higher TIR (and greater emphasis on text) likely increases attentional competition between text and pictorial content. Thus, introducing more eye-catching text to posts with information-rich pictorial messages may distract and detract from their efficacy.

We also consider the presence of people and social scenes (i.e., social content) as a moderator and separate category of pictorial content. Pictorial content with people (and their faces) opens the opportunity for many social cognition-based persuasion strategies. For example, marketers may hire well-known or trustworthy-looking endorsers to provide social proof (Cialdini and Goldstein 2004), and pictures or videos of smiling actors can provide positive social signaling (Yin, Jia, and Zheng 2021). Posts with more social content may also rely less on verbal reasoning or text-based information.

Content congruency. Finally, we test whether congruency between text and pictorial content moderates the effect of TIR. One possibility is that greater congruence between the information content of text and picture helps reinforce the intended message and improves processing fluency, which should enhance attitude formation and engagement (Ceylan, Diehl, and Proserpio 2024; Edell and Staelin 1983). An alternative possibility is that greater congruence provides redundant information that is less engaging and receives less cognitive processing (Houston, Childers, and Heckler 1987).

Data and Processing

Data

To test the relationship between TIR and social media engagement across different visual aesthetic and content contexts in MGC, we collected two large field datasets of organic posts ($N_{Facebook} = 25{,}466$, $N_{Instagram} = 27{,}498$) from 100 official brand accounts on Facebook and 95 official brand accounts on Instagram that spanned 69 brands across 9 major product categories.[2] To determine the coverage of brands and products, we used public rankings such as Forbes's brand ranking to choose the popular brands within each product category (see Web Appendix B for a full list of brands). We subsequently collected the most recent 500 organic posts from each account; the Instagram posts were collected before July 2021, and the Facebook posts were collected before August 2022.[3] Some

[2] We identified at least two different business accounts for each of the listed brands on each platform: the accounts for the same brand may differ in language (most are in English or Chinese) or region (worldwide or country-based).

[3] We focus on the posts with images and excluded posts without any image or those containing other multimedia, such as short videos.

Table 3. Definition and Operationalization of Constructs.

Construct	Definition	Operationalization
Dependent Variable		
Engagement	Primary measures of social media engagement	Continuous; total number of likes or comments each post received
Text–Picture Balance		
Text–image ratio (TIR)	Proportion of surface space allocated to text content within a post image (0%–100%)	Continuous; proportion of total surface space on an image allocated to the text content detected by Google Vision AI
Spatial Arrangement of Text		
Centeredness	Spatial arrangement of text elements on a post image relative to the image center	Continuous; the distance between the center of mass for all the text blocks and the center of the image
Information Value of Text		
Text entropy	Informativeness of overlay text based on the number of information entities extracted from text	Continuous; the random entropy of a given text content $E = \log N$, where N is the number of distinct entities (e.g., date, price, location) detected by Google Natural Language AI
Text sentiment	Sentiment score of overlay text	Continuous; ranging from −1 (negative) to 1 (positive), detected by Google Natural Language AI
Marketing mix (robustness check)	The presence or count of marketing-mix information on overlay text (i.e., 4Ps: product, price, place, and promotion)	Binary (presence) and continuous (count); presence: coded as 1 when 4Ps-related entities are detected; count: the sum of 4Ps-related entities detected
Information Value of Picture		
Visual entropy	Informativeness of pictorial content based on the number of objects detected from image	Continuous; the random entropy of pictorial content $E = \log N$, where N is the number of objects (e.g., persons, product images, and dominant objects in the background) detected by Google Vision AI (the max. number of objects the algorithm returns is 10)
Social content	Presence of human faces	Binary; coded as 1 when a post image contains one or more faces detected by Google Vision AI
Content Congruency		
Text–picture congruency	Congruency (semantic similarity) between the pictorial and text content in a post image	Continuous; an index formulated by the cosine similarity between the pictorial and text elements in a post image using CLIP
Visual Aesthetics		
Logo	Presence of logos and other brand elements	Binary; coded as 1 when a post image contains one or more logos or brand elements detected by Google Vision AI
Visual complexity	The complexity of a post's basic visual features, color, luminance, and edges	Continuous; logarithm of image file size (in kilobytes)
Colorfulness	The amount of color variation in a post image	Continuous; the pixel percentage of the top three colors detected by Google Vision AI: the lower the pixel percentage of the colors, the more colorful the post image
Color hue	The dominant hue of a post including cold, warm, black and white, or no dominant color	Categorical; dummy-coded variables for the color hues in a post: neutral, cold, warm, and mixed. Mixed serves as the reference group
Brightness	Overall brightness of a post image	Continuous; average value across all image pixels (HSV color representation model)
Brightness contrast	The difference in brightness between adjacent or nearby pixels	Continuous; standard deviation of value across all image pixels
Caption Content		
Caption: entropy	Informativeness of caption text based on the number of information entities extracted from caption text	Continuous; the random entropy of a given text content $E = \log N$, where N is the number of distinct entities (e.g., date, price, location) detected by Google Natural Language AI
Caption: sentiment	Sentiment score of caption text	Continuous; ranging from −1 (negative) to 1 (positive), detected by Google Natural Language AI

less-active accounts had fewer than 500 posts; we excluded inactive accounts (which might not be comparable), which we define as accounts with fewer than 50 posts in total since establishment. We also excluded posts that were published within ten days of the date of data collection (see Web Appendix C for details about the collection and cleaning process). Finally, we focused on posts that contain overlay text (72% of Facebook and 62% of Instagram posts). This yielded the final sample for our main analyses with 17,914 posts from Facebook and 16,696 posts from Instagram. Although our brand coverage is not exhaustive, the diversity and breadth of our data are a significant advance on previously published research, which has

tended to focus on one or two product categories (e.g., Li and Xie 2020) and UGC as opposed to MGC (e.g., Hartmann et al. 2021; Li and Xie 2020).

We used several industry-leading computer vision and NLP tools to process the large-scale unstructured social media data and extract content-related variables of interest (see Table 2 for a summary of the data processing tools and the variables extracted). These tools enabled us to isolate and classify specific elements of social media posts such as text content, objects, brand logos, and human faces as well as gestalt visual aesthetic features such as color profile, brightness, visual complexity, and layout (see Web Appendix D for more details and examples). Compared to manual or human-subjects-derived coding of visual information, computer vision tools enable researchers to process large datasets in an objective, nonbiased, and efficient manner.

We primarily used Google Vision AI and Google Natural Language AI (industry-leading computer vision and NLP techniques developed by Google available at https://cloud.google.com) to process the unstructured image and text data, respectively. In addition, we used specialized tools such as CLIP, an OpenAI neural network algorithm that can predict the most relevant text for an image, to measure congruency between text and pictorial content (https://openai.com/research/clip). Next, we describe the variables and how they were extracted using these tools (see Table 3 for the definition and operationalization of each variable and Table 4 for summary statistics).

Variables

Dependent variables. We follow standard industry practice of operationalizing social media engagement as likes and comments, which are two of the most basic and common metrics of social media engagement (Peters et al. 2013). Previous research measuring social media engagement has typically operationalized the dependent variable using likes, shares, or comments, depending on the social media platform (e.g., Hartmann et al. 2021; Lee, Hosanagar, and Nair 2018). We use likes and comments as separate dependent variables; we do not consider shares because it is only publicly visible on Facebook and not on Instagram.

Likes, the simplest form of consumer engagement on social media platforms, are "one click" reactions to posts that enable users to express interest, enjoyment, or endorsement of the content without much effort and deliberation. The number of likes a post receives provides marketers with a simple and intuitive metric of audience reaction to social media content (Mochon et al. 2017). Comments are predominantly text-based reactions to social media posts or responses to other users. These reactions are considered to be a higher level of engagement because writing content in response to posted content demands greater cognitive involvement and effort from users (Cho, Schweickart, and Haase 2014).

Overall, likes capture simple valanced reactions, such as affective reaction, whereas comments are cognitively triggered and relate to more rational or interactive features of social media content (Kim and Yang 2017). All the aforementioned measurements of social media engagement have managerial consequences: Previous research has linked likes, shares, and comments to marketing key performance indicators including sales, customer acquisition, brand community membership, and loyalty program participation (Liadeli, Sotgiu, and Verlegh 2023; Mochon et al. 2017).

Text–image ratio. To operationalize the relative balance between text and picture in the post image, we calculate the proportion of surface area allocated to text on an image, which we call TIR. We measure TIR by employing the Optical Character Recognition (OCR) algorithm of Google Vision AI, which automatically detects text elements in multiple colors, irregular fonts, and complex layouts from cluttered backgrounds (see Web Appendix D). After identifying the presence of text blocks, the detector returns their coordinates, which we use to compute an aggregated estimate for text space and calculate the proportion of surface space allocated to text content, that is, TIR.[4] We observe significant heterogeneity in TIR in MGC across different brands (Figure 4), which suggests a lack of consensus on best practice or that the effect of TIR is context dependent.

Moderator 1: Text centeredness. Text centeredness indicates the spatial arrangement of text elements on a post image relative to the image center (details in Web Appendix E). To determine text centeredness, we first calculate the center of mass for overlay text using the coordinates of all detected text elements, weighted by the number of words each text element contains. We then calculate the distance of this text center of mass from the image center (0,0) and normalize this distance based on the length between the image's corner and center (i.e., half of the diagonal length).

Moderator 2a: Text entropy. To assess the information value of text, we consider two text content factors: text entropy and text sentiment. Text entropy measures the information content of the overlay text based on Shannon entropy. Entropy is a measure of information value that is theoretically grounded in information theory (and the favored measure of information quantity in fields such as computer science and mathematics; see Shannon 1948). To measure text entropy, we first use the entity analysis tool in Google Natural Language AI to detect entities (e.g., consumer goods, price, location) in the overlay text (see Web Appendix D).[5] We then compute the random entropy of a given text, $\text{Entropy} = \log N$, where N is the number of distinct entities. As an alternative measure of

[4] Post images may contain more than one text block, such as clusters of same-style text elements. Their spatial arrangement is captured by centeredness, the distance of the center of mass of all the blocks from the image center (see Web Appendix E).

[5] There are 11 types of entities prespecified by Google Natural Language AI: DATE, CONSUMER_GOOD, PRICE, NUMBER, PERSON, LOCATION, ORGANIZATION, EVENT, WORK_OF_ART, ADDRESS, and PHONE_NUMBER. See Google Cloud (2025) for more information.

Table 4. Summary Statistics.

	Facebook (N = 17,914)				Instagram (N = 16,696)			
	Mean	**SD**	**Min**	**Max**	**Mean**	**SD**	**Min**	**Max**
Social Media Engagement								
No. of likes	1,335	11,154	0	470,000	19,009	63,384	0	2,262,000
No. of comments	62	519	0	43,000	118	456	0	15,000
Text–Picture Balance								
Text–image ratio (TIR)	.11	.12	0	1	.09	.12	0	1
Moderators								
Overlay text: centeredness	−.32	.20	−.94	0	−.35	.2	−.96	0
Overlay text: entropy	1.15	.95	0	3.46	.92	.91	0	3.32
Overlay text: sentiment score	.21	.32	−.95	.98	.15	.3	−.96	.98
Overlay text: sentiment magnitude	.53	.70	0	17.77	.44	.71	0	20.45
Visual entropy	1.06	.81	0	2.30	1.05	.8	0	2.3
Social content: face presence[a]	.40	.49	0	1	.38	.49	0	1
Text–picture congruency	.29	.08	0	.52	.26	.07	.07	.52
Marketing-mix info: presence[a]	.37	.48	0	1	.3	.46	0	1
Marketing-mix info: count	.53	.82	0	4	.38	.67	0	4
Caption Content								
Caption: entropy	1.95	.90	0	3.46	1.86	.9	0	3.46
Caption: sentiment	.44	.28	−.90	.98	.42	.27	−.91	.98
Visual Aesthetics								
Logo presence[a]	.58	.49	0	1	.43	.5	0	1
Visual complexity	3.65	.76	−.11	6.77	4.22	.45	1.22	5.89
Colorfulness	.57	.23	0	.99	.56	.22	0	.98
Color hues:								
Warm hue[a]	.29	.46	0	1	.27	.44	0	1
Cold hue[a]	.29	.45	0	1	.32	.47	0	1
Mixed hue[a]	.34	.47	0	1	.33	.47	0	1
Neutral hue[a]	.08	.27	0	1	.08	.28	0	1
Brightness	.45	.11	.01	.93	.43	.11	0	.91
Contrast	.32	.06	.07	.49	.31	.05	.05	.48
Posting Time								
Weekend dummy[a]	.16	.36	0	1	.29	.45	0	1
Post duration (days)	748	729	11	3,366	597	565	11	3,163

[a]Indicates dummy variable.

information content, we distinguish the marketing-mix entities (i.e., consumer goods, price, location, and promotion) from other entities and consider their presence and count (i.e., information amount) as a robustness check (see Web Appendix F for correlations between variables).

Moderator 2b: Text sentiment. We use the sentiment analysis tool in Google Natural Language AI to assess the sentiment of overlay text, which was extracted using the OCR detector. The language model assigns a sentiment score to the text, ranging from −1.0 (highly negative) to +1.0 (highly positive) with the midpoint (.0) indicating neutrality.

Moderator 3a: Visual entropy. To operationalize the information value of a post's pictorial content, we calculate the visual entropy of a post image (Lee and Lee 2015). Similar to text entropy, we employ Shannon entropy to capture the amount of information conveyed by pictorial objects in the image. We first use object detection tools in Google Vision AI to detect objects (e.g., persons, products, and dominant objects in the background) in the image. We then use the same formula, Entropy = logN, to compute the random entropy of a given post image where N counts the objects detected.

Moderator 3b: Social content in image. We also consider the presence of human faces in an image, which we use as a proxy measure for social content in an image. We use Google Vision AI's facial detection feature, which detects faces in an image and the associated key facial attributes (including emotional state).

Moderator 4: Text–picture congruency. We use a neural network algorithm developed by OpenAI, CLIP, to compute an index measure of the congruency between overlay text and pictorial content on the image. This measure is based on the semantic similarity (as calculated in cosine similarity) of different pairs of the pictorial and text elements detected in a post image.

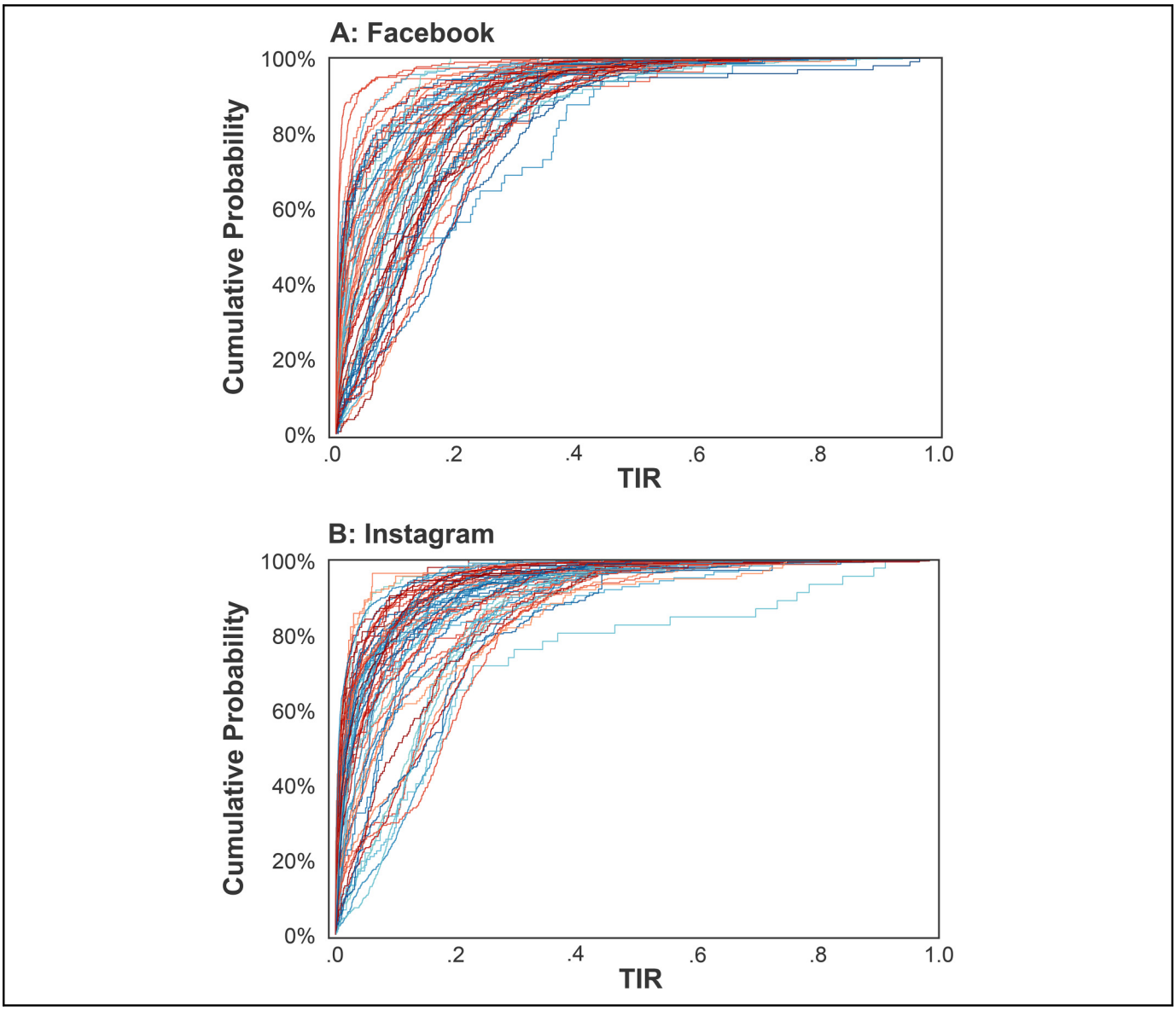


Figure 4. Brand-Level Cumulative Distributions of TIR.
Notes: Cumulative distribution of TIR for each post of a given brand on each platform (each curve represents a different brand).

Controls. As established in the literature (Hartmann et al. 2021; Li and Xie 2020), we use two types of controls: post-related controls and visual aesthetic controls. The former includes time-related factors such as posting time, date, and duration; account-level factors that capture observed differences across accounts such as number of followers and account establishment duration; and account fixed effects, which capture the unobserved differences such as account objectives, posting frequency, and follower profiles. The latter covers a diversified set of aesthetic configurations such as colorfulness, color hues, brightness, and brightness contrast (Li and Xie 2020). We further consider the effect of visual complexity given its relevance to attention and comprehensibility (Pieters, Wedel, and Batra 2010; Rosenholz, Li, and Nakano 2007).

Empirical Model

The empirical analysis has two main objectives: (1) to test the effect of TIR on social media engagement and (2) to quantify the heterogeneous treatment effects of TIR across various content and visual context moderators, including spatial arrangement of text, information value of text, information value of picture, and content congruency. We employ CCR-CF, a machine learning-based causal inference method

Table 5. Comparison of Models.

	Estimable Effects		Statistical Properties	Continuous Treatment Compatibility		Functional Form Flexibility	Ability to Address Endogeneity Arising from Unobserved Factors	Robustness
Model	ATE	HTE	Asymptotic Guarantees and Confidence Intervals	Continuous Treatment Without Heterogeneous Intensity	Continuous Treatment with Heterogeneous Intensity	Relax the Parametric Functional Form Assumption	Relax the Unconfoundedness Assumption	Dual Robustness
OLS	✓	✗	✓	✓	✗	✗	✗	✗
PSM + NegB	✓	✗	✓	✗	✗	✗	✗	✗
Two-stage least squares	✓	✗	✓	✓	✗	✗	✓	✗
Generalized causal tree	✓	✓	✗	✓	✗	✓	✗	✗
CIVF	✓	✓	✓	✓	✓	✓	✓	✗
CCR-CF	✓	✓	✓	✓	✓	✓	✗	✓

Notes: CCR-CF and CIVF are the models used in this article. ATE refers to average treatment effect, and HTE means whether the model can estimate individual-level heterogeneous treatment effects. Treatment with heterogeneous intensity means that the treatment may depend on the value of covariates. PSM + NegB refers to models that employ a negative binomial model to analyze the relationship between a count variable and other covariates, following matching with the traditional PSM method.

under the generalized random forests (GRF) framework (Athey, Tibshirani, and Wager 2019). As an extension of GRF, CCR-CF offers robustness to observed confounders and clustered data.

CCR-CF has several features that allow for more reliable and robust estimation of heterogeneous treatment effects (i.e., how the effect of TIR varies across content and visual contexts), especially when the relationship between the outcome variable and the covariates is complex (Athey, Tibshirani, and Wager 2019; Wager and Athey 2018), as is the case for our data. First, the term "confounding-robust" reflects the model's ability to mitigate bias from misspecification of observed confounders by flexibly modeling the relationship among covariates, treatment, and outcome. Traditional parametric methods (e.g., ordinary least squares [OLS], propensity score matching [PSM], or negative binomial models) often fail to capture nonlinear or high-dimensional confounding structures in the data, leading to residual bias even when all confounders are observed.[6] CCR-CF addresses this by adaptively partitioning covariate space to isolate subgroups where treatment assignment approximates randomness within those subgroups (see Web Appendix G).

Second, "cluster-robust" refers to CCR-CF's capacity to handle dependencies in clustered data (e.g., each account stands as one cluster in our case), ensuring the estimates are not biased by within-cluster correlations. Confounding- and cluster-robustness ensures valid inference of heterogeneous treatment effects in the presence of complex confounding structures among observed covariates and when dealing with clustered data. Last, CCR-CF can be used in contexts where a continuous treatment variable exhibits heterogeneous treatment intensity, which is the case for TIR, a continuous variable that may be influenced by post and page characteristics.

Researchers in marketing, computer science, and medical fields have employed causal forest–based models in contexts where the treatment variables are continuous (Feuerriegel et al. 2024; Tabib and Larocque 2024; Wan et al. 2022; Zhang and Luo 2023). Other emerging methods, such as the generalized causal tree developed by Nandy et al. (2023), can also handle continuous treatment variables but were developed for policy learning and lack asymptotic guarantees and confidence intervals for their estimates. Table 5 compares the methodological properties of CCR-CF with alternative approaches.

An important assumption made by CCR-CF is unconfoundedness. This means that the treatment assignment W_i is as good as random, conditional on the observed covariates X_i (Rosenbaum and Rubin 1983). Subsequently, we relax this assumption in a robustness check to address unobserved confounding by allowing for an instrumental variable (IV) in the causal forest, a method known as Causal IV Forests (CIVF), which is an application of the GRF method (Athey,

[6] CCR-CF cannot address unconfoundedness violations but excels in handling misspecified confounder relationships. Unlike parametric models, it is less biased when X's effect has unknown or complex functional forms.

Tibshirani, and Wager 2019). Another important assumption is the overlap assumption, which requires that each unit has a nonzero probability of receiving any treatment level, ensuring that treatment assignment is not deterministic based on covariates. CCR-CF also complies with SUTVA (the stable unit treatment value assumption) and assumes smooth variation of treatment effects with covariates (regularity condition).

In summary, CCR-CF distinguishes itself by enabling nonparametric estimation of heterogeneous treatment effects under continuous treatments with heterogeneous intensity while providing asymptotic guarantees and cluster-robust inference. Unlike OLS, PSM, or two-stage least squares—which require parametric specifications—CCR-CF avoids functional form restrictions and is robust to complex observed confounding factors. It also accommodates clustered dependencies and heterogeneous treatment intensity without assuming constant marginal effects. In addition, we conducted a series of Monte Carlo simulations to test the performance of the CCR-CF model with a continuous treatment variable and various complex relationships between covariates and outcomes (see Web Appendix H). Next, we outline the CCR-CF model specification and discuss potential concerns regarding endogeneity as well as the conditions under which CCR-CF can mitigate these issues.

CCR-CF Model

The CCR-CF model consists of an outcome variable Y (in our case, the number of likes and comments), a treatment variable W (TIR), and a set of covariates X (i.e., features) that include moderators of interest M and control variables. The relationship between them is modeled as follows (Athey, Tibshirani, and Wager 2019):

$$Y_i = W_i \cdot \theta(X_i) + q(X_i) + \varepsilon_i . \quad (1)$$

The goal is to estimate $\hat{\theta}(x)$, which is the conditional average treatment effect (CATE) of TIR, that is, the average treatment effect of TIR for different levels of the moderating variables. The average treatment effect (i.e., main effect) of TIR, $\bar{\theta}$, is obtained by integrating CATE over X. The function q is a flexible function of moderators and other controls, which is estimated nonparametrically using flexible machine learning methods. One of the most desirable properties of CCR-CF is that it can obtain a point estimate of the treatment effect for each observation, which enables us to obtain a multidimensional distribution of θ as a function of X.

At a high level, CCR-CF can be thought of as fitting a forest to solve the local moment conditions: $E\left[(Y - W(x) \cdot \hat{\theta}(x) - q(x) | X = x\right] = 0$. In a forest with B total number of trees, each node in the tree b contains a local estimate of the parameter, $\hat{\theta}_b(x)$, for every region of X that falls within that leaf. Intuitively, the forest aggregates these estimates across all trees through weighted combinations to produce a more robust global parameter estimate $\hat{\theta}_{forest}(x)$. The tree-splitting criterion is to find splits that maximize the heterogeneity score, which is used to measure the degree of heterogeneity in the treatment effects. Web Appendix G provides a more detailed description of the CCR-CF model, including the estimation steps and key assumptions. Next, we describe the main steps for estimating the CCR-CF model.

Step 1 (Orthogonalization). We orthogonalize Y_i and W_i by transforming them into $\tilde{Y}_i$ and $\tilde{W}_i$, where $\tilde{Y}_i = Y_i - \hat{y}^{-i}(X_i)$, and $\tilde{W}_i = W_i - \hat{w}^{-i}(X_i)$. Specifically, $\hat{y}^{-i}(\cdot)$ and $\hat{w}^{-i}(\cdot)$ are leave-one-out estimates of the marginal expectations using flexible machine learning algorithms, computed without using the ith observation. CCR-CF with orthogonalization better addresses cases of both confounding and treatment heterogeneity compared with the methods in the original causal forest model by Wager and Athey (2018). This shares the same spirit with the doubly robust machine learning approach of Chernozhukov et al. (2018).

Step 2 (Grow Cluster-Robust Random Forest). We then grow the random forest with honest trees to get $\alpha_i(x)$, where $\alpha_i(x)$ is a data-adaptive kernel that measures how often the ith training example falls in the same leaf as the test point x.

$$\alpha_i(x) = \frac{1}{B}\sum_{h=1}^{B} \frac{1(\{X_i \in L_b(x),\ i \in S_b\})}{|\{i{:}X_i \in L_b(x),\ i \in S_b\}|}.$$

$L_b(x)$ denotes the leaf of the bth tree containing the training sample x. Random forests make predictions as an average of b trees. First, for each tree $b = 1, \ldots, B$, a subsample $S_b \in \{1, \ldots, n\}$ is drawn. Then, a tree grows via recursive partitioning on each subsample of the data: We make predictions by aggregating these estimates across all trees (Athey and Wager 2019; Athey, Tibshirani, and Wager 2019). We also allow for page-level clustering to account for potential correlations within each cluster. Rather than directly drawing a subsample of observations, we draw a subsample of clusters $J_b \in \{1, \ldots, J\}$, where J is the total number of pages, when growing the trees. Then, S_b is generated by drawing samples at random from each cluster $j \in J_b$.

Step 3 (Estimate CATE). Following Athey and Wager (2019), we plug in the kernel weights $\alpha_i(x)$ obtained in step 2 to get the CATE estimate $\hat{\theta}(x)$:

$$\hat{\theta}(x) = \frac{\sum_{i=1}^{n} \alpha_i(x)\left(Y_i - \hat{y}^{-i}(X_i)\right)\left(W_i - \hat{w}^{-i}(X_i)\right)}{\sum_{i=1}^{n} \alpha_i(x)\left(W_i - \hat{w}^{-i}(X_i)\right)^2},$$

which is the covariance between $\tilde{Y}_i$ and $\tilde{W}_i$ divided by the variance of $\tilde{W}_i$. That is, $\hat{\theta}(x) = \frac{\sum_{b=1}^{B} \mathrm{Cov}\left[\tilde{Y}_i, \tilde{W}_i \mid X=x\right]}{\sum_{b=1}^{B} \mathrm{Var}\left[\tilde{W}_i \mid X=x\right]}$. After we get the point estimate $\hat{\theta}(x)$ for each sample i in our data, we can examine the heterogeneous treatment effects (i.e., moderation)

Table 6. Average and Heterogeneous Treatment Effects of TIR.

	Facebook		Instagram	
Dependent Variable	**Likes**	**Comments**	**Likes**	**Comments**
Average Treatment Effect				
Text–image ratio (TIR)	−2.390	−1.170	−5.215	−3.616
	(.408)	(.372)	(.598)	(.452)
	[.000]	[.002]	[.000]	[.000]
Heterogeneous Treatment Effect				
Spatial arrangement of text				
TIR × Text centeredness	2.070	1.376	4.646	3.541
	(.025)	(.017)	(.031)	(.024)
	[.000]	[.000]	[.000]	[.000]
Information value of text				
TIR × Text entropy	.393	.247	.645	.396
	(.006)	(.004)	(.010)	(.008)
	[.000]	[.000]	[.000]	[.000]
TIR × Text sentiment	1.006	.569	.937	.694
	(.018)	(.012)	(.032)	(.024)
	[.000]	[.000]	[.000]	[.000]
Information value of picture				
TIR × Visual entropy	−.238	−.172	−.499	−.441
	(.007)	(.005)	(.012)	(.009)
	[.000]	[.000]	[.000]	[.000]
TIR × Social content: face presence	−.278	−.235	−.599	−.510
	(.013)	(.008)	(.021)	(.016)
	[.000]	[.000]	[.000]	[.000]
Content congruency				
TIR × Text–picture congruency	7.506	4.789	10.735	6.983
	(.057)	(.039)	(.134)	(.107)
	[.000]	[.000]	[.000]	[.000]
N	17,914		16,696	

Notes: Standard errors are in parentheses, and *p*-values are in brackets.

of each proposed moderator by regressing $\hat{\theta}(x)$ on the moderator M using simple linear regression:

$$\hat{\theta}(x_i) = \alpha + M_i\varphi + \xi_i. \quad (2)$$

For example, if M is the centeredness of overlay text, then φ captures how the effect of TIR changes at different levels of text centeredness; this can be interpreted as the interaction effect between TIR and centeredness. We then repeat the same analysis for other moderators. We use regression to present heterogeneous treatment effects in a manner comparable with traditional statistical models. Subsequently, we fully quantify and visualize the CATE estimates.

Concerns over Endogeneity

Endogeneity concerns are inherent in causal inference from field data. Here we discuss the possible sources of endogeneity in our investigation and explain how we address them. First, TIR and other visual and content features may be the result of strategic choices by brands to enhance engagement, resulting in selection bias. For example, marketers might naturally opt for a larger TIR when content is more informative. Second, post engagement is concurrently influenced by various observed factors, such as aesthetic factors (e.g., colorfulness, brightness, contrast), caption features (e.g., entropy, sentiment), and time-related variables (e.g., posting time, duration). Inadequate control of these factors can lead to biased estimates of treatment effects. These two concerns stem from observable confounding factors, which CCR-CF can address under the unconfoundedness conditional on observables assumption. Our analysis controls for various covariates, including post-level content features, page characteristics, and time-related controls like post duration and fixed effects for weekends and months. Additionally, biases may arise from unobservable factors, such as time-variant unobserved post or account characteristics that affect the choice of TIR as well as the post's engagement and virality. We address this concern by relaxing the unconfoundedness assumption and running a robustness check with the CIVF model, which we discuss in the "Robustness Checks" section.

Results

This section is organized in four parts. First, we present the primary analysis on the average treatment effect of TIR on social media engagement. Second, we test how the TIR effect may depend on the information content and visual aesthetics of the post by examining the four categories of moderators. Third,

we delineate how the effect of text–picture balance varies across different levels of moderators. Last, we conduct a series of robustness tests with alternative model specifications to demonstrate the reliability of our results and rule out alternative explanations.

Main Effect of TIR on Engagement

Since our research question is how the balance of text and picture affects engagement, our primary analysis only includes social media posts with overlay text (72% of Facebook and 62% of Instagram posts). We subsequently conduct a robustness check using samples that include posts without overlay text. Here and in later models, the estimate $\bar{\theta}$, the average treatment effect of TIR on engagement metrics, is computed after controlling for X, which includes all the moderators of interest (i.e., text centeredness, text entropy, text sentiment, visual entropy, social content, and text–picture congruency) as well as control variables (i.e., caption entropy, caption sentiment, visual complexity, colorfulness, color hue, brightness, brightness contrast, post duration, day-of-the-week, and month fixed effects of the post). Table 6 summarizes the main effect of TIR and its interaction with context moderators.

TIR has a significant negative effect on social media engagement, as measured by likes and comments. Overall, this effect is consistent with earlier research demonstrating "picture superiority" in marketing communication (Li and Xie 2020; Pieters and Wedel 2004). There are noticeable differences across social media platforms: The effect of TIR is less negative for Facebook ($\theta = -2.39$ and -1.17 for likes and comments, respectively) than for Instagram ($\theta = -5.22$ and -3.62, respectively), which is consistent with the notion that Instagram is a pictorially oriented content platform, whereas Facebook is relatively more text oriented. There are also differences between engagement metrics; the coefficient is less negative for comments than for likes, which implies that a greater emphasis on text is less disadvantageous for eliciting comments, which are often text-based responses.

Moderating Factors

Next, we test whether the effect of TIR changes across the four content and visual contexts we previously discussed (Figure 3). We test for moderating effects by estimating the heterogeneous treatment effects of TIR across these contexts.

Spatial arrangement of text. Text centeredness has a significant positive effect on likes and comments on both platforms ($ps = .000$). Although moving the text to the center of a post image is a simple way to boost user engagement for text-oriented posts, there may be caveats. Our results could also reflect that successful posts centered a parsimonious quantum of text, which may have focused on the most compelling and interesting information (e.g., the core brand message, sales information, or a catchy tagline). However, centering a large quantity of text without regard for parsimony may create clutter that impedes visual attentional processes (Pieters, Wedel, and Zhang 2007).

Information value of text. We examine two aspects of the information value of the text content in posts: text entropy and text sentiment. Previous research often uses word count as a measure of informativeness (e.g., Zhang and Luo 2023). Here, we use Shannon entropy of text as an alternative measure of information quantity, because raw word count may be endogenous with TIR (a higher word count typically means that the text occupies more space) and conflated with readability (a higher word count may denote greater wordiness rather than more information). To avoid multicollinearity, we do not include both entropy and word count in our analyses.

The negative impact of TIR on engagement weakens as text entropy and text sentiment increase ($ps = .000$). In other words, higher TIR is not always detrimental for engagement, particularly when the text conveys more information and positive sentiment. Notably, the positive moderation of text entropy is more pronounced on Instagram (likes: .65 vs. .39 for Instagram and Facebook; comments: .40 vs. .25, respectively). This could be because users on a primarily pictorial platform like Instagram tend to omit essential text information, and a higher TIR mitigates this tendency.

Information value of picture. The negative effect of TIR on engagement is significantly amplified by visual entropy and social content, which have significant negative effects ($ps = .000$). Higher TIR is even more detrimental for engagement when pictorial content is information rich, for example, when the image contains many individual objects in the background or depicts a social setting (where human faces are present). In particular, human faces may indicate the presence of influence agents and denote the use of social proof strategies in a post (Cialdini and Goldstein 2004; Yin, Jia, and Zheng 2021). The negative moderation suggests that a higher TIR fosters competition between text and pictorial information, hindering one's ability to efficiently process the pictorial content.

Content congruency. The effect of TIR becomes less negative when the pictorial and text content are more congruent. We find significant heterogeneous effects of TIR across text–picture congruency levels ($ps = .000$); text–picture congruency is a highly significant moderator with a relatively large positive parameter value. In other words, posts that emphasize text (i.e., higher TIR) perform better if the text content is congruent and consistent with the pictorial content. It is possible that posts perform better when both text and picture are used to support and reinforce a core message or particularly compelling information. Expressing the same information both verbally and pictorially may also improve processing fluency and increase the likelihood that consumers comprehend a message (Van Rompay, De Vries, and Van Venrooij 2010). This can also be achieved via attention capture and transfer (Pieters and Wedel 2004), where consumer attention is first captured by the pictorial content before shifting to the text content. Such positive effects overwhelm any negative effects of information redundancy on user engagement.

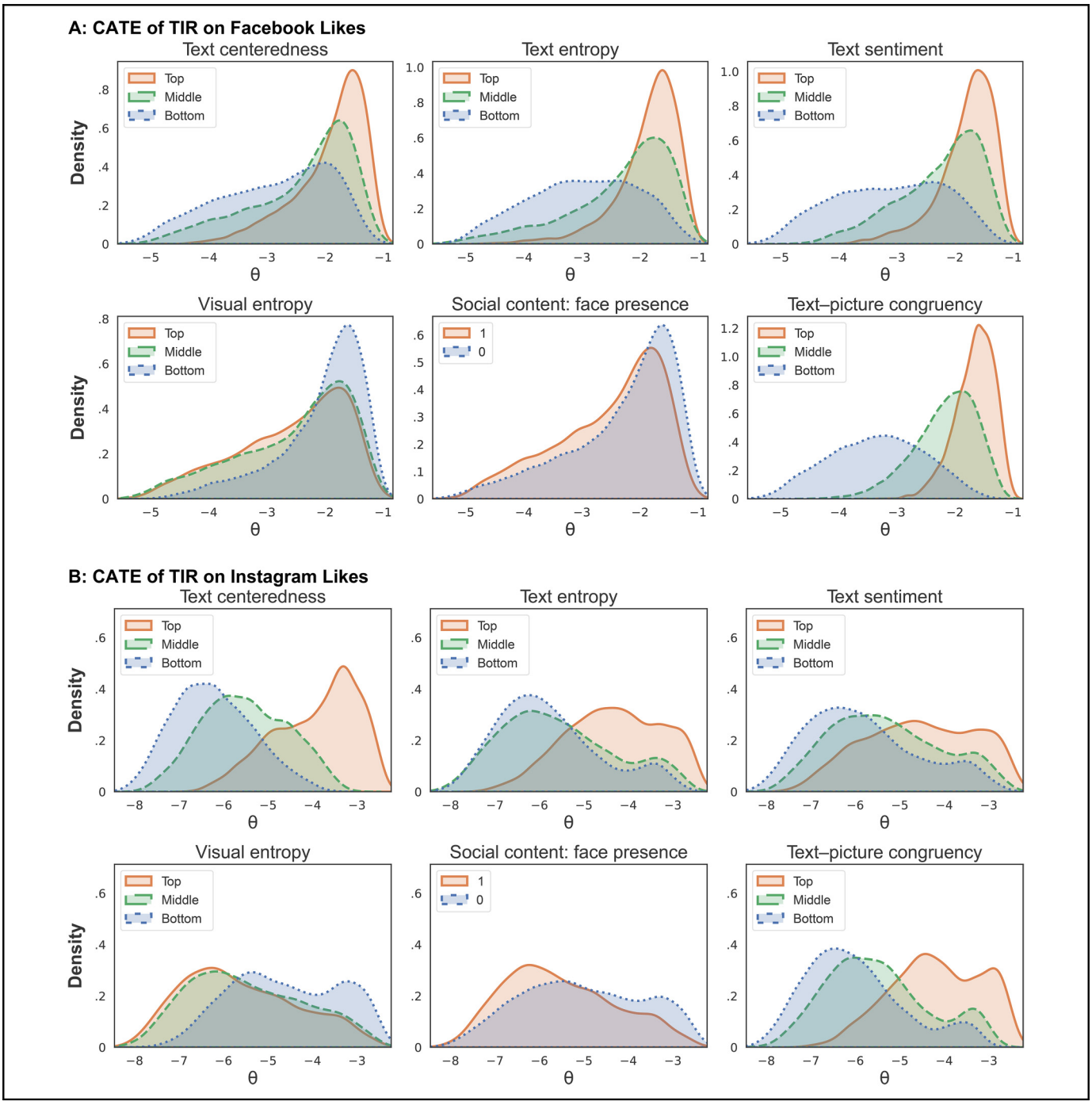


Figure 5. Heterogeneous Treatment Effects of TIR at Different Moderator Levels.

Notes: Each line represents the distribution of the CATE of TIR (i.e., θ) at a given level of moderator. For continuous-value moderators, the sample is divided into three groups: top (posts with the top one-third moderator values), middle (posts with middle one-third moderator values), and bottom (posts with bottom one-third moderator values). For binary-value moderators, the sample is divided into two groups: posts with moderator values equal to 1 and posts with moderator values equal to 0.

Heterogeneity and Incremental Changes Across Moderator Levels

The CCR-CF model can estimate the CATE of TIR, the average treatment effect of TIR for different subsamples of posts with similar features (Athey, Tibshirani, and Wager 2019). We exploit this feature to examine how the effect of TIR varies at different levels of moderator variables. This provides a more nuanced exposition of the empirical effects, showing how much incremental changes in specific visual or content features affect social media engagement.

Conditional average treatment effects (CATE). Panels A and B of Figure 5 visualize the heterogeneous treatment effects of TIR on likes when content and visual context moderators vary in level. Each plot focuses on one moderator. First, we rank the sample based on the value of each moderator. Then, we divide the sample into three groups for each continuous-value moderator: posts with the top one-third moderator values, posts with the middle one-third moderator values, and posts with the bottom one-third moderator values. Next, we plot the distribution of CATE estimates for these three groups. For the presence of faces, which is a binary-value moderator, we divide the sample into posts with and without faces. A wider and flatter distribution in these density plots indicates greater heterogeneity in the treatment effect, whereas a distribution leaning toward the left-hand side suggests a more negative effect. Greater overlap between distributions for a particular moderator represents a weaker moderation effect.

To check that these effects are not spurious statistical results, we follow Athey and Wager's (2019) heuristic approach to test for heterogeneity in the TIR effects. We split the sample into three subsamples—top, middle, and bottom one-third—based on individual TIR estimates and repeat the CCR-CF estimation for each. As shown in Figure G1 of Web Appendix G, the results reveal significant heterogeneity in TIR effects. This underscores the importance of a model that excels in estimating heterogeneous treatment effects.

On both Facebook and Instagram, the effect of TIR becomes progressively less negative when the text is more centered, contains more information, evokes more positive affect, or is more congruent with pictorial content. Text-related moderators (i.e., text centeredness, text entropy, text sentiment, text–picture congruency) also exhibit greater heterogeneity compared with pictorial moderators (i.e., visual entropy and social content), which have a weak (and negative) moderation effect. We repeat the analysis for comments and obtain similar results (Web Appendix I).

There are noticeable differences across platforms: There is less overlap and greater heterogeneity in the CATE distributions for Instagram compared with Facebook, particularly for text-related moderators. This could reflect that Instagram is a visual-storytelling platform where users are more sensitive and fickler about feature quality (like the arrangement of text and snappier writing), compared with Facebook users, who may be more tolerant of wordy posts.

Marginal effect of TIR at different levels of moderators. In Web Appendix J, we also compute how the effect of TIR varies at these different levels of moderating variables. Specifically, we compute the marginal effect of TIR at one standard deviation above the mean, the mean, and one standard deviation below the mean of each moderator.

Robustness Checks

Alternative models. First, we conduct a confounding-robust causal forest model without clustering. The results are largely the same with the CCR-CF model (Web Appendix K). Second, to address potential unobserved confounding factors that may bias the results, such as time-variant manager learning, we employ the CIVF model, which uses IVs in causal forests. Specifically, we use the historical TIR of peer pages on the other platform (Instagram or Facebook) as the IV for the focal page's current TIR. We construct the IV as follows: First, we assess page similarity based on various characteristics, creating a similarity score between the focal page and all other pages in the same category. We then identify the top three most similar pages on the other platform, referred to as peer pages. Finally, we calculate the average TIR of historical posts from these peer pages, using this historical average as the IV for the current TIR of the focal page. The idea is that historical TIR of peer pages reflects industry standards, which the focal page may learn from (relevancy condition). However, such historical industry norms are unrelated to the focal brand's current unobserved learning curve (exclusion condition). We also check the robustness of the CIVF model by redefining the peer page as the three most similar pages from the same platform. We discuss the two IVs and results in detail in Web Appendix K.

Including posts without overlay text. Our primary analyses exclude posts without overlay text. As a robustness check, we repeat the same analyses using the unconditional sample that includes posts with and without overlay text (Web Appendix L). The findings remain qualitatively the same as the analyses that excluded posts without overlay text.

Excluding posts with TIR below specific thresholds. The high precision of computer vision tools may detect small text elements embedded in other objects such as logos or clothing, which may bias the estimation of the TIR effects. To validate the reliability of computer vision for text size estimation, we conduct robustness analyses using data that exclude posts with TIR below 1% or 5% and obtain consistent results (Web Appendix M).

Text–picture balance and marketing-mix information. In the primary analysis, we use text entropy as a measure of information value of text. As a robustness check that also has managerial relevance, we use the amount of marketing-mix information (i.e., product, price, promotion, and place) as an alternative operationalization of the information value of text. We repeat the main analysis using the presence and quantity of marketing-mix information (i.e., marketing-mix presence and count, respectively) in overlay text as the two measures of "information value of text" and replicate our previous findings (Web Appendix N).

Alternative measure of text–picture balance. In the main analysis, we operationalized text–picture balance using TIR, which is a relative concept and measure. To test whether the effect of TIR depends on the absolute size of either element, we replace TIR with text size (surface area of text) and use picture size (surface area of pictorial content) as a control variable in a robustness check. As additional robustness checks, we also rerun the main model using two different subsamples:

one selecting for posts with similar text sizes but varying picture sizes, and the other selecting for posts with similar picture sizes but varying text sizes. All robustness checks yield the same pattern of results (same signed coefficients for TIR and moderators) as our primary results (Web Appendix O).

Controlling for the size and position of objects and logos. The main model accounts for the presence of logos and visual entropy but does not consider their sizes or spatial arrangement. To address the possibility that the relative size of logos and pictorial elements and their spatial arrangement may influence the results, we conduct robustness checks by including both the sizes and the centeredness of pictorial objects and logos as controls; we obtain consistent results (Web Appendix P).

Alternative measure of text sentiment. Google Natural Language AI offers two sentiment values. Our primary investigation uses the Google sentiment score—a value ranging from −1 (negative) to +1 (positive), which reflects the overall sentiment polarity (valence)—to gauge text sentiment. For robustness, we also consider an additional sentiment value from Google Natural Language AI, which measures the "magnitude" of sentiment by capturing the strength of emotion within the text, regardless of positivity or negativity, and ranges from 0 to positive infinity. We obtain consistent results when using either magnitude or sentiment, or even if both parameters are included (Web Appendix Q).

Product and brand type. We explore potential heterogeneous effects for products versus services and also for hedonic versus utilitarian brands by conducting independent analyses for each. These checks examine the heterogeneity caused by specific classification variables on top of the interaction between TIR and specific moderators and account for unobservable feature differences across categories and brand types. We observe similar results (i.e., CATE distributions) across different classifications (see Web Appendix R).

Managerial Guidelines

Social media has become one of the most vital and competitive channels for marketing communications, accounting for 17% of total marketing budgets (The CMO Survey 2023). Understanding and quantifying how specific content and visual features drive engagement in MGC has immense practical value. We focus on how the relative emphasis of two fundamental communication elements, text and picture, affects engagement. We summarize our empirical results into two overarching sets of managerial guidelines: (1) identifying content- and context-dependent design principles (i.e., "it depends") and (2) quantifying the impact of design features (i.e., "how much"). These insights can help marketers move beyond simplistic heuristics toward a quantitative, adaptive, automation-ready approach to visual content strategy.

"It Depends…"

Although, holding other factors constant, a larger size of overlay text relative to the image reduces engagement, there are several steps managers can take to reduce this negative impact. We also conduct a robustness check that calculates how the mere presence of text in a post affects engagement while controlling for visual and content features. We find that the mere presence of text has no statistically significant effect for both Facebook and Instagram posts (Web Appendix L).

Rather, our research reveals that there is no one-size-fits-all approach. The optimal text–picture balance is shaped by content and visual aesthetics and is highly context dependent. For example, while wordy posts generally receive less engagement, posts containing important text-based information—which may require more words—can perform better. This is true when the text fits naturally within the image and aligns with the overall visual and content context. By examining the key conceptual moderators (i.e., heterogeneous treatment effects), our study provides a nuanced set of guidelines that inform managers when to enlarge or shrink overlay text, as summarized in Figure 6. With these insights, managers can counteract the potential drawbacks of emphasizing text and maximize the joint impact of text and pictorial content on social media engagement.

When text complements picture or is essential for conveying a clear and compelling message (whether informative or emotional), marketers should emphasize text. Enlarging and centering text is beneficial when the text contains important information (e.g., key marketing-mix information including product and price; see also Web Appendix N), evokes strong positive emotions, or aligns with the message conveyed within the image. Conversely, shrinking text is more engaging when pictorial content is already rich in detail and speaks on its own. Adding or emphasizing text can be distracting or even detracting when the image contains prominent focal objects like key products or social cues like human faces.

In addition to answering how to align text with visuals and content, this research offers insights into whether, where, and when to use overlay text on post images. These contextual considerations are summarized in Table 7 as a set of practical rules of thumb. First, managers should use overlay text judiciously. For complex messages and calls to action, overlay text is often essential and should not be avoided. Second, managers should consider platform differences: Emphasizing text tends to be less effective on pictorially oriented platforms like Instagram but may elicit greater engagement on more text-oriented platforms like Facebook. Furthermore, engagement goals also matter. While minimizing text can help grab initial attention (and more likes), more emphasis on text could help drive discussion and text-based interactions in the comments.

Together, these highly contingent guidelines reinforce the key point that the optimal text–picture balance also depends on visuals, content, and context. Marketers need to move beyond simplistic heuristics, such as "minimizing text" or following a magic number like Facebook's former 20% text overlay rule. Indeed, overlay text can improve engagement if it is strategically placed and sized appropriately for specific content and context.

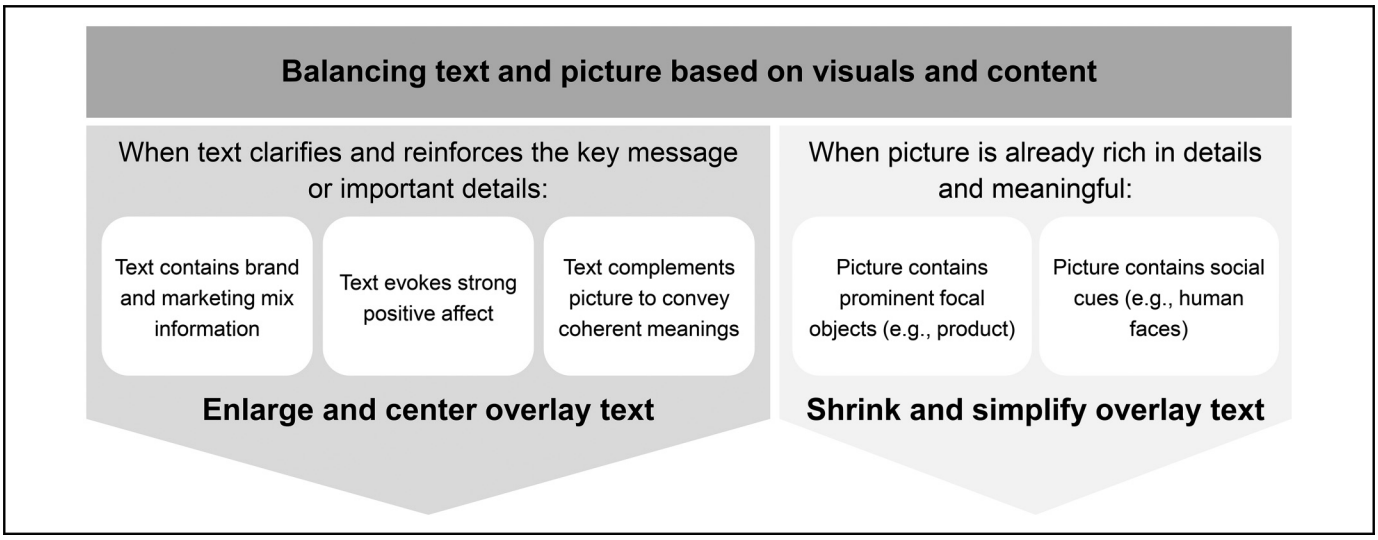


Figure 6. Optimizing Text–Picture Balance: Visual and Content Considerations.

Marketers must harmonize the visual design with the intended message of the content in a given context. Our research also offers a framework for testing and quantifying how much individual content elements and visual aesthetics affect social media engagement, which we analyze next to answer the "how much" question.

"How Much"

The open-ended nature of social media content allows for infinite design possibilities: Marketers must decide what mix of text content, pictorial content, and visual aesthetic features (e.g., color, tones, and structure) can yield the best engagement. Rather than relying on intuition, marketers can now use off-the-shelf toolkits to measure and quantify how specific features shape engagement. We leverage the causal machine learning model to address two basic content design questions and discuss their strategic implications.

Feature priority. A fundamental question in MGC design is "What content and visual features should one prioritize?" To answer this, we computed the feature importance score for each moderating variable, which quantifies the contribution of individual features to the variability in the effect of our focal treatment variable (TIR) across different observations. Figure 7 ranks all context moderators based on their feature importance. We find that the most influential factors for engagement are text centeredness and text–picture congruency, followed by text sentiment; this is consistent across platforms. In contrast, entropy measures (both text and visual entropy) exhibit lower importance. While the presence of human faces (social content) has a statistically significant effect, its influence on engagement is minimal.

We observe differences across platforms: Text centeredness has the highest importance on Instagram, whereas text–picture congruency and text sentiment are relatively more important on Facebook. It is possible that on pictorially oriented platforms like Instagram, aesthetic considerations such as the spatial position of text matter more, whereas on relatively text-heavy platforms like Facebook, the substance of text (e.g., whether text conveys emotion, how text complements picture) matters more.

Table 7. A Rule-of-Thumb Guide for Balancing Text and Picture in MGC Across Contexts.

Guideline	Key Insights	Application
Whether: Use overlay text judiciously	Overlay text works best when picture alone cannot fully convey the message. Use it to clarify or reinforce meaning.	Use text for abstract pictorial content, complex messages, or key information like product explanations, promotions, and calls to action.
Where: Configure text based on platform	Text should be more prominent on information-rich platforms and minimized on pictorially oriented platforms.	Different formatting depending on platform: – Facebook: larger, centered text – Instagram: smaller, simplified text
When: Tailor to engagement goals	Salient text can encourage comments (conversation) but may reduce likes (attention).	Enlarge and center text for dialogue-driven campaigns; minimize text and let picture speak for attention-focused posts.
How: Align text with visuals and content	Overlay text should harmonize with both design (visual) and message (content).	Configure text strategically based on the guidelines in Figure 6.

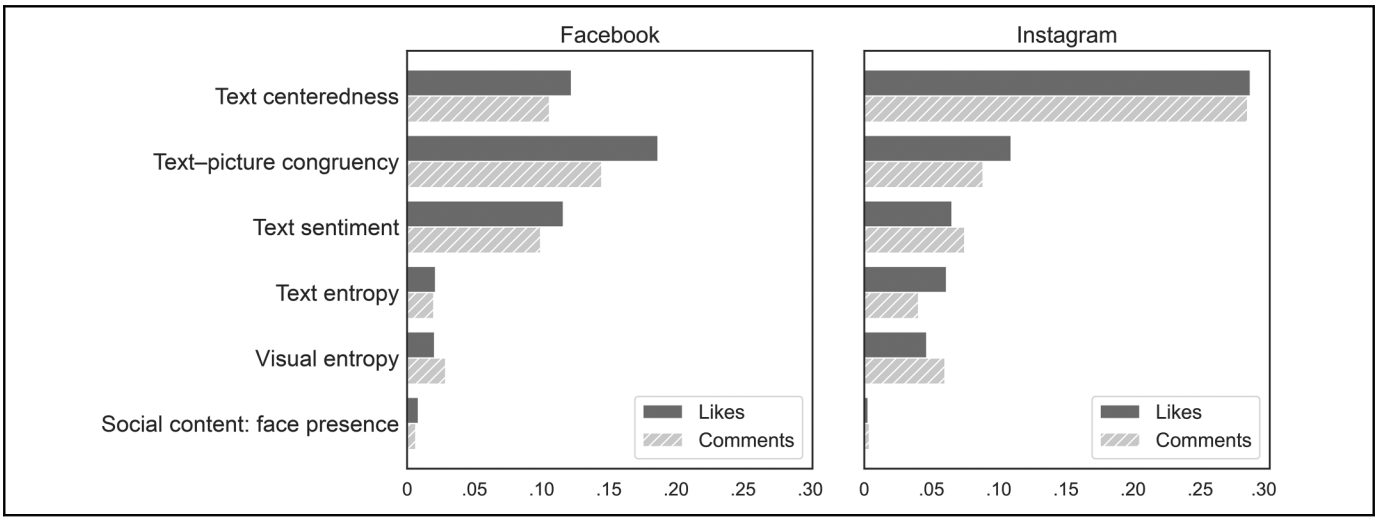


Figure 7. Comparing the Importance of Content and Visual Features.
Notes: The feature importance score is calculated based on the amount of parameter heterogeneity it creates. The higher the value, the more important the feature is in terms of its contribution to the heterogeneous treatment effects of the focal treatment variable, TIR.

Content and visual elasticities. The CCR-CF model also enables us to estimate CATE at different levels of a specific content and visual feature (or set of features), which quantifies how much engagement shifts in response to incremental changes in these features. We refer to these estimations as "elasticities of social media engagement." Distinct from feature importance, which assesses how much a given feature contributes to the overall heterogeneity in treatment effects, elasticities capture the extent to which engagement varies as content and visual features change.[7]

Table 8 summarizes the complete set of content and visual elasticities. These estimates are based on a typical post where text covers 10% of the image's surface area (i.e., the mean value of TIR in our sample).[8] By comparing the percentage changes in engagement for a unit change (i.e., one standard deviation) in visual or content variables, we obtain valuable quantitative insights for MGC design. On Instagram, for instance, centering larger text can increase likes and comments by 7.60% and 3.36%, respectively. Additionally, a unit increase in text entropy leads to a 2.52% increase in likes and a 4.86% increase in comments. In contrast, a unit increase in visual entropy decreases engagement by 6.18% (likes) and 5.05% (comments). Congruent content improves engagement by 5.63% (likes) and 6.46% (comments).

General Discussion

Marketers have long debated the relative virtue of using semantic (text) versus pictorial representations, which are the two most basic forms of communications. However, there is little empirical understanding of how emphasizing one versus the other affects engagement. We use computer vision and NLP tools to extract the text and pictorial content and visual aesthetic features of 51,881 organic brand posts from Facebook and Instagram, and examine how these content and visual features correspond to actual social media engagement. Central to our investigation is how the relative balance of text and picture (operationalized by TIR) affects engagement as measured by likes and comments.

Overall, we find that TIR has a significant negative effect on engagement: MGC that emphasizes text over picture receives lower social media engagement. This main effect is robust across all models for both Facebook and Instagram. This implies that overall, text and picture have a competitive rather than complementary relationship. However, the direction and extent of this effect depend on the visual and content features of the post. In particular, the negative effect of TIR is positively moderated by the spatial arrangement of text, information value of text, information value of picture, and content congruency. Posts perform better when overlay text is more centered and when text and pictorial content are congruent; both factors might improve processing fluency (Wyer, Hung, and Jiang 2008). Posts receive more engagement when text contains more information, as measured by text entropy or marketing-mix information;

[7] The elasticity analysis shows how moderators affect engagement. Notably, the direction of these changes may differ from the interaction effects observed in the main analysis (Table 6), where the signs indicate how content moderators alter the impact of TIR.

[8] Because causal forests do not directly facilitate counterfactual prediction, we reconstruct predicted outcomes using the formula $\hat{Y} = \hat{y}(X_i) + \theta(X_i) \times (W - \hat{w}(X_i)) + \bar{\epsilon}$, where $\hat{y}(X_i)$ is the outcome model prediction and $\hat{w}(X_i)$ denotes the treatment model prediction (both $\hat{y}(X_i)$ and $\hat{w}(X_i)$ are used in the orthogonalization step of the CCR-CF model estimation) $\theta(X_i)$ is the estimated heterogeneous treatment effect, and $\bar{\epsilon}$ is the mean residual across the data sample. For each feature of interest, we generate predictions for both the typical sample (using mean values) and a counterfactual scenario where the feature is increased by one standard deviation. The percentage change is calculated using the formula $100\% \times (\hat{Y}_{\text{counterfactual}} - \hat{Y}_{\text{typical}})/(\hat{Y}_{\text{typical}})$.

Table 8. Visual and Content Elasticities of Engagement.

	Facebook		Instagram	
	Δ No. of Likes	Δ No. of Comments	Δ No. of Likes	Δ No. of Comments
Spatial Arrangement				
Text centeredness	5.54%	2.69%	7.60%	3.36%
Information Value of Text				
Text entropy[a]	4.02%	2.89%	2.52%	4.86%
Text sentiment	.74%	.65%	1.29%	−5.03%
Information Value of Picture				
Visual entropy	−3.27%	−2.32%	−6.18%	−5.05%
Social content: face presence	.13%	.09%	−.09%	−.07%
Content Congruency				
Text–picture congruency	5.69%	2.34%	5.63%	6.46%

[a]We use the count of the marketing-mix information as a measure of text entropy in this table to generate easy-to-implement managerial implications.
Notes: This table presents the percentage changes in engagement corresponding to a unit change in visual and content moderators for a typical post on Facebook and Instagram. A one-unit change in a variable indicates an increase by one standard deviation from its mean. We employ a counterfactual prediction approach using the CCR-CF model to estimate the percentage change in outcomes when feature values increase from the mean to one standard deviation above the mean. Notably, the direction of these changes may differ from the interaction effects observed in the main analysis (Table 6), where the signs indicate how content moderators alter the impact of TIR.

intuitively, consumers should value more useful information and marketing-mix information in MGC. Posts also perform better when text contains emotional content: It is possible that emotional content evokes more episodic or visual imagery in consumers' minds, which is relatively more complementary with visual information (MacInnis and Price 1987).

We also identify negative moderators: factors that may aggravate the negative effect of TIR. Relatively more text-oriented posts receive lower engagement when the picture contains more information (higher visual entropy) or depicts social scenes where human faces are present, such as in posts where human models are providing social proof. In both cases, more pictorially represented information might increase the redundancy of text-based information; thus, dedicating more space to pictorial content may be best when the picture speaks for itself.

Conceptual Implications

Empirically modeling the effect of text–picture balance on social media engagement contributes to the visual marketing, content design, and communication literature. Previous research studying content design and consumer engagement with social media has largely focused on the effect of visual aesthetic factors—such as color, brightness, and clarity—or content in isolation. For example, Li and Xie (2020) examine the effect of pictorial content on social media engagement for UGC and show that the mere presence of an image yields more engagement (compared with a post with no image). In practice, however, marketers and content creators do not face a binary decision but need to optimize the balance of both picture and text within an image. Despite the ubiquity of both text and picture in MGC (as well as in UGC and other media), relatively little is known about how they jointly affect consumer engagement and response. We show that TIR has a significant negative effect on engagement, which suggests that text and picture, despite individually being indispensable, may coexist in tension. The findings—that relatively larger text (or relatively smaller "picture") corresponds to less social media engagement—is consistent with the long-standing wisdom that pictorial representations are better at winning customer attention than text, that is, "the picture superiority effect" (e.g., Childers and Houston 1984; Edell and Staelin 1983; Rossiter and Percy 1997; Shepard 1967). Whereas previous literature focused on differences in how consumers process text versus pictorial information, we quantify how the balance of text and picture (as a continuous variable) affects consumer engagement.

It is worth noting that TIR is a stronger and more robust predictor of engagement than all the other visual and content aesthetic factors in our study. The effect is significant for all models, for different dependent variables (likes or comments), across platforms, and robust even after controlling for a myriad of visual aesthetic factors or when pooling over all Facebook and Instagram data together (Web Appendix S). This suggests that the construct captures features that are fundamental to marketing communications and plays a major role in eliciting consumer engagement to MGC.

The significant moderation effects also show that the precise impact of TIR depends on information content and visual aesthetic design. These effects highlight the limitations in following an absolute "picture superiority" heuristic in practice. Ultimately, picture cannot fully replace text in many contexts, and marketers cannot simply avoid using overlay text in MGC when their goal is to convey information. Rather, marketing communications depends on achieving the right balance between text and picture, which we find also depends on the spatial arrangement of text, information value of text, information value of picture, and content congruency. The moderators suggest that information content and visual aesthetics jointly interact to determine the marketing efficacy of MGC, a

perspective that has received relatively little attention in the empirical visual marketing literature.

Our focus on MGC further differentiates our results from other research using computer vision tools to link the visual properties of social media posts to consumer response, which primarily examine UGC (e.g., Dzyabura, El Kihal, and Peres 2022; Hartmann et al. 2021; Li and Xie 2020; Liu, Dzyabura, and Mizik 2020). There are significant differences between UGC and MGC: MGC is designed to further a firm's marketing goals by delivering advertising content, communicating with consumers, and building deeper relationships with followers (Kumar et al. 2016; Meire et al. 2019; Tellis et al. 2019). Our research also differs from previous research on MGC, which has focused primarily on how text content, independent of visual properties, affects engagement (e.g., Goh, Heng, and Lin 2013; Lee, Hosanagar, and Nair 2018; see Table 1).

Limitations and Future Research

Balancing text and picture is a fundamental part of marketing communications and content design, but there is surprisingly little empirical understanding of how the joint use of these communication elements affects consumer response. In taking a first step in quantifying this question, we inevitably leave many questions unanswered. For example, we focus on MGC content by larger brands, which typically have professional social media managers, to obtain a dataset of content that is more likely to represent contemporary best practices. Although our models control for account-level differences such as popularity, our analysis does not cover content by smaller brands or private businesses, which may be more similar to content produced by consumers. Whether the effects we documented extend to smaller brands, professional influencers, and consumers (i.e., UGC) is an open question. Given that the interaction norms between posters and consumers are different for each of these categories, we would not be surprised if there were significant differences. Nevertheless, the constructs we tested may readily be extended to other contexts to test completely different hypotheses.

Future research can consider how text–picture balance affects other measures of social media engagement as well as other downstream outcome variables. In this research, we used counts of likes and comments as dependent variables, partly because they are well-established measures of engagement but also because of data collection constraints. For example, alternative measures of engagement such as views or clicks are not publicly visible on Facebook and Instagram, while measures such as shares are only visible on Facebook but not on Instagram. As an exploratory analysis, we repeated our main analysis using Facebook shares as the dependent variable and found consistent results and the same signs for the moderators; however, TIR has a marginally significant main effect (Web Appendix T). This might reflect that text–picture balance has a stronger effect in eliciting attention and response (in the forms of likes and comments) but that sharing (and word of mouth) should be considered a separate category of behavior (at least for Facebook).

Future research can also explore additional moderators and sources of heterogeneity for our effect. For example, we find that cross-category variations for the moderation effects are generally more pronounced on Facebook than on Instagram, consistent with the potential platform differences (in relative text vs. pictorial orientation) discussed in the main text. Demonstrating the precise reason for such cross-category differences may require more sophisticated categorization of post content, which should be possible as off-the-shelf machine vision tools become increasingly sophisticated. It should be noted that the accuracy of our calculations depends on the number of control variables; whereas we controlled for dozens of variables, future researchers may be able to control for hundreds of variables or more.

Beyond Heuristics: Toward a Structured Adaptive Approach to Optimizing MGC

Overall, our research contributes to a growing literature that leverages advanced image and text processing tools, alongside large unstructured datasets of social media posts, to quantify the complex interplay between text and pictorial content in shaping actual consumer engagement. Although progress is still needed before marketers can establish a fully generalized algorithm for optimizing MGC or visual aesthetics, what was once mystified as an imprecise art is increasingly evolving into a data-driven science. While traditional guidelines—such as the universal applicability of "picture superiority"—have been valuable heuristics, they may not consistently hold across the multitudinous possibilities of content and context.

As advances in generative AI reduce the cost of marketing content creation, the attention-economy of social media will grow even more competitive. To stay ahead of competition, practitioners must adopt content strategies that are adaptive, evidence based, and contextually relevant. The ability to identify, specify, and quantify the optimal content features for distinct contexts (i.e., figuring out "it depends" rules and also quantifying "how much") may give marketers, advertisers, and content creators an edge in this highly competitive and ever-evolving landscape. Future content creation models may integrate generative AI, feature testing, and iterative feedback from real-time social media engagement, transforming a practice once guided by intuition and trial and error into a structured process of automation and feature optimization.

Acknowledgments

The authors would like to express their sincere gratitude to the *JMR* review team for their constructive comments and insightful suggestions, which have greatly enhanced the quality of this article. The authors are also grateful to Xiufan Yu and Mengxia Zhang for their valuable feedback. Additionally, they appreciate the comments from Michael He Jia, Robert S. Wyer Jr., Bennett Yim, Austin Zhuang, and participants at the Association for Consumer Research Conference 2020 and Marketing Science Conference 2021. The authors also gratefully acknowledge the outstanding research assistance from Zhili Li and Yu Ye. The authors used ChatGPT 4.1 for proofreading.

Author Contributions

Ivy Chu Dang and Canice M.C. Kwan contributed equally and are listed in alphabetical order. Correspondence should be directed to them.

Coeditor

Rebecca Hamilton

Associate Editor

Anindita Chakravarty

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by the Research Grant Council of Hong Kong (grant nos. C7105-20G and GRF-17506824), the Basic Research Fund from HKU Business School Shenzhen Research Institutes (grant no. SZRI2023-BRF-05), and the National Natural Science Foundation of China (grant nos. 72422010, 71702200, 71702201, and 72472104).

ORCID iDs

Ivy Chu Dang https://orcid.org/0000-0001-6349-6320
Canice M.C. Kwan https://orcid.org/0000-0002-7127-6122
Jayson S. Jia https://orcid.org/0000-0002-3645-6857
Yang Shi https://orcid.org/0000-0002-5156-1858

References

Ahmadi, Iman, Adrian Waltenrath, and Christian Janze (2023), "Congruency and Users' Sharing on Social Media Platforms: A Novel Approach for Analyzing Content," *Journal of Advertising*, 52 (3), 369–86.

Athey, Susan, Julie Tibshirani, and Stefan Wager (2019), "Generalized Random Forests," *Annals of Statistics*, 47 (2), 1148–78.

Athey, Susan and Stefan Wager (2019), "Estimating Treatment Effects with Causal Forests: An Application," *Observational Studies*, 5 (2), 37–51.

Berger, Jonah and Katherine L. Milkman (2012), "What Makes Online Content Viral?" *Journal of Marketing Research*, 49 (2), 192–205.

Ceylan, Gizem, Kristin Diehl, and Davide Proserpio (2024), "Words Meet Photos: When and Why Photos Increase Review Helpfulness," *Journal of Marketing Research*, 61 (1), 5–26.

Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins (2018), "Double/Debiased Machine Learning for Treatment and Structural Parameters," *The Econometrics Journal*, 21 (1), 1–68.

Childers, Terry L. and Michael J. Houston (1984), "Conditions for a Picture Superiority Effect on Consumer Memory," *Journal of Consumer Research*, 11 (2), 643–54.

Cho, Moonhee, Tiffany Schweickart, and Abigail Haase (2014), "Public Engagement with Nonprofit Organizations on Facebook," *Public Relations Review*, 40 (3), 565–67.

Cialdini, Robert B. and Noah J. Goldstein (2004), "Social Influence: Compliance and Conformity," *Annual Review of Psychology*, 55, 591–621.

The CMO Survey (2023), "Managing Brand, Growth, and Metrics," (March), https://cmosurvey.org/wp-content/uploads/2023/03/The_CMO_Survey-Highlights_and_Insights_Report-March_2023.pdf.

Dzyabura, Daria, Siham El Kihal, and Renana Peres (2022), "Image Analytics in Marketing," in *Handbook of Market Research*, Christian Homburg, Martin Klarmann, and Arnd Vomberg, eds. Springer International Publishing, 665–92.

Edell, Julie A. and Richard Staelin (1983), "The Information Processing of Pictures in Print Advertisements," *Journal of Consumer Research*, 10 (1), 45–61.

Feuerriegel, Stefan, Dennis Frauen, Valentyn Melnychuk, Jonas Schweisthal, Konstantin Hess, Alicia Curth, Stefan Bauer, Niki Kilbertus, Isaac S. Kohane, and Mihaela van der Schaar (2024), "Causal Machine Learning for Predicting Treatment Outcomes," *Nature Medicine*, 30 (4), 958–68.

Goh, Khim-Yong, Cheng-Suang Heng, and Zhijie Lin (2013), "Social Media Brand Community and Consumer Behavior: Quantifying the Relative Impact of User- and Marketer-Generated Content," *Information Systems Research*, 24 (1), 88–107.

Google Cloud (2025), "Entity," Cloud Natural Language API (accessed October 15, 2025), https://cloud.google.com/natural-language/docs/reference/rest/v1/Entity#Type.

Hartmann, Jochen, Mark Heitmann, Christina Schamp, and Oded Netzer (2021), "The Power of Brand Selfies," *Journal of Marketing Research*, 58 (6), 1159–77.

He, Jiaxiu, Bingqing Li, and Xin Shane Wang (2023), "Image Features and Demand in the Sharing Economy: A Study of Airbnb," *International Journal of Research in Marketing*, 40 (4), 760–80.

Houston, Michael J., Terry L. Childers, and Susan E. Heckler (1987), "Picture-Word Consistency and the Elaborative Processing of Advertisements," *Journal of Marketing Research*, 24 (4), 359–69.

Ivins, William M., Jr. (1969), *Prints and Visual Communication*. MIT Press.

Jalali, Nima Y. and Purushottam Papatla (2019), "Composing Tweets to Increase Retweets," *International Journal of Research in Marketing*, 36 (4), 647–68.

James, Henry (1893), *Picture and Text*. Harper and Brothers.

Kim, Cheonsoo and Sung-Un Yang (2017), "Like, Comment, and Share on Facebook: How Each Behavior Differs from the Other," *Public Relations Review*, 43 (2), 441–49.

Kübler, Raoul V., Lara Lobschat, Lina Welke, and Hugo van der Meij (2024), "The Effect of Review Images on Review Helpfulness: A Contingency Approach," *Journal of Retailing*, 100 (1), 5–23.

Kumar, Ashish, Ram Bezawada, Rishika Rishika, Ramkumar Janakiraman, and P.K. Kannan (2016), "From Social to Sale: The Effects of Firm-Generated Content in Social Media on Customer Behavior," *Journal of Marketing*, 80 (1), 7–25.

Kwan, Canice M. C., Xianchi Dai, and Robert S. Wyer Jr. (2017), "Contextual Influences on Message Persuasion: The Effect of Empty Space," *Journal of Consumer Research*, 44 (2), 448–64.

Lee, Dokyun, Kartik Hosanagar, and Harikesh S. Nair (2018), "Advertising Content and Consumer Engagement on Social Media: Evidence from Facebook," *Management Science*, 64 (11), 5105–31.

Lee, Kwanghyun and Sanghoon Lee (2015), "A New Framework for Measuring 2D and 3D Visual Information in Terms of Entropy," *IEEE Transactions on Circuits and Systems for Video Technology*, 26 (11), 2015–27.

Li, Yiyi and Ying Xie (2020), "Is a Picture Worth a Thousand Words? An Empirical Study of Image Content and Social Media Engagement," *Journal of Marketing Research*, 57 (1), 1–19.

Liadeli, Georgia, Francesca Sotgiu, and Peeter W.J. Verlegh (2023), "A Meta-Analysis of the Effects of Brands' Owned Social Media on Social Media Engagement and Sales," *Journal of Marketing*, 87 (3), 406–27.

Liu, Liu, Daria Dzyabura, and Natalie Mizik (2020), "Visual Listening in: Extracting Brand Image Portrayed on Social Media," *Marketing Science*, 39 (4), 669–86.

Lu, Shijie, Isaac Dinner, and Rajdeep Grewal (2023), "The Ripple Effect of Firm-Generated Content on New Movie Releases," *Journal of Marketing Research*, 60 (5), 908–31.

MacInnis, Deborah J. and Linda L. Price (1987), "The Role of Imagery in Information Processing: Review and Extensions," *Journal of Consumer Research*, 13 (4), 473–91.

Meire, Matthijs, Kelly Hewett, Michel Ballings, V. Kumar, and Dirk Van den Poel (2019), "The Role of Marketer-Generated Content in Customer Engagement Marketing," *Journal of Marketing*, 83 (6), 21–42.

Mochon, Daniel, Karen Johnson, Janet Schwartz, and Dan Ariely (2017), "What Are Likes Worth? A Facebook Page Field Experiment," *Journal of Marketing Research*, 54 (2), 306–17.

Nandy, Preetam, Xiufan Yu, Wanjun Liu, Ye Tu, Kinjal Basu, and Shaunak Chatterjee (2023), "Generalized Causal Tree for Uplift Modeling," in *2023 IEEE International Conference on Big Data (BigData)*. Institute of Electrical and Electronics Engineers, 788–98.

Overgoor, Gijs, William Rand, Willemijn van Dolen, and Masoud Mazloom (2022), "Simplicity Is Not Key: Understanding Firm-Generated Social Media Images and Consumer Liking," *International Journal of Research in Marketing*, 39 (3), 639–55.

Peters, Kay, Yubo Chen, Andreas M. Kaplan, Björn Ognibeni, and Koen Pauwels (2013), "Social Media Metrics—A Framework and Guidelines for Managing Social Media," *Journal of Interactive Marketing*, 27 (4), 281–98.

Pieters, Rik and Michel Wedel (2004), "Attention Capture and Transfer in Advertising: Brand, Pictorial, and Text-Size Effect," *Journal of Marketing*, 68 (2), 36–50.

Pieters, Rik, Michel Wedel, and Rajeev Batra (2010), "The Stopping Power of Advertising: Measures and Effects of Visual Complexity," *Journal of Marketing*, 74 (5), 48–60.

Pieters, Rik, Michel Wedel, and Jie Zhang (2007), "Optimal Feature Advertising Design Under Competitive Clutter," *Management Science*, 53 (11), 1815–28.

Pillai, Kishore Gopalakrishna, Constantine S. Katsikeas, and Caterina Presi (2012), "Print Advertising: Type Size Effects," *Journal of Business Research*, 65 (6), 865–68.

Rayner, Keith, Caren M. Rotello, Andrew J. Stewart, and Jessica Keir (2001), "Integrating Text and Pictorial Information: Eye Movements When Looking at Print Advertisements," *Journal of Experimental Psychology: Applied*, 7 (3), 219–26.

Rosenbaum, Paul R. and Donald B. Rubin (1983), "The Central Role of the Propensity Score in Observational Studies for Causal Effects," *Biometrika*, 70 (1), 41–55.

Rosenholz, Ruth, Yuanzhen Li, and Lisa Nakano (2007), "Measuring Visual Clutter," *Journal of Vision*, 7 (2), 1–22.

Rossiter, John R. and Larry Percy (1997), *Advertising Communications and Promotion Management*, 2nd ed. McGraw-Hill.

Shannon, Claude E. (1948), "A Mathematical Theory of Communication," *Bell System Technical Journal*, 27 (3), 379–423.

Shepard, Roger N. (1967), "Recognition Memory for Words, Sentences, and Pictures," *Journal of Verbal Learning and Verbal Behavior*, 6 (1), 156–63.

Smith, Ruth Ann (1991), "The Effects of Visual and Verbal Advertising Information on Consumers' Inferences," *Journal of Advertising*, 20 (4), 13–24.

Stafford, Marla Royne (1996), "Tangibility in Services Advertising: An Investigation of Verbal Versus Visual Cues," *Journal of Advertising*, 25 (3), 13–28.

Tabib, Sami and Denis Larocque (2024), "Comparison of Random Forest Methods for Conditional Average Treatment Effect Estimation with a Continuous Treatment," *Statistical Methods in Medical Research*, 33 (11–12), 1952–66.

Tellis, Gerard J., Deborah J. MacInnis, Seshadri Tirunillai, and Yanwei Zhang (2019), "What Drives Virality (Sharing) of Online Digital Content? The Critical Role of Information, Emotion, and Brand Prominence," *Journal of Marketing*, 83 (4), 1–20.

Van Rompay, Thomas J.L., Peter W. De Vries, and Xaviera G. Van Venrooij (2010), "More Than Words: On the Importance of Picture–Text Congruence in the Online Environment," *Journal of Interactive Marketing*, 24 (1), 22–30.

Wager, Stefan and Susan Athey (2018), "Estimation and Inference of Heterogeneous Treatment Effects Using Random Forests," *Journal of the American Statistical Association*, 113 (523), 1228–42.

Wan, Shu, Chen Zheng, Zhonggen Sun, Mengfan Xu, Xiaoqing Yang, Hongtu Zhu, and Jiecheng Guo (2022), "GCF: Generalized Causal Forest for Heterogeneous Treatment Effect Estimation in Online Marketplace," arXiv, https://doi.org/10.48550/arXiv.2203.10975.

Wedel, Michel and Rik Pieters (2017), "A Review of Eye-Tracking Research in Marketing," *Review of Marketing Research*, 4, 123–47.

Wyer, Robert S., Jr., Iris W. Hung, and Yuwei Jiang (2008), "Visual and Verbal Processing Strategies in Comprehension and Judgment," *Journal of Consumer Psychology*, 18 (4), 244–57.

Yin, Yunlu, Jayson S. Jia, and Wanyi Zheng (2021), "The Effect of Slow-Motion Video on Consumer Inference," *Journal of Marketing Research*, 58 (5), 1007–24.

Zhang, Mengxia and Lan Luo (2023), "Can Consumer-Posted Photos Serve as a Leading Indicator of Restaurant Survival? Evidence from Yelp," *Management Science*, 69 (1), 25–50.

Copyright of Journal of Marketing Research (JMR) is the property of American Marketing Association and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.